Round-Optimal Identity-Based Blind Signature from Module Lattice Assumptions

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Abstract

This work presents a round optimal identity-based blind signature scheme based on module lattices. Our construction extends Fischlin's two-round blind signature framework [CRYPTO'06] to the identity-based setting. The construction uses the GPV signature scheme based on Micciancio and Peikert's G-trapdoor techniques and the widely adopted framework of Lyubashevsky, Nguyen and Plançon [CRYPTO'22] in the random-oracle model. The scheme is secure under the MLWE and MSIS assumptions. The optimised parameters are also provided targeting 128-bit security. To the best of our knowledge, this scheme is the first identity-based blind signature scheme which relies on module lattice problems and is round-optimal.

1 Introduction

Blind signatures are a crucial component of modern cryptography. It is used in numerous real-world applications such as cryptocurrencies, e-voting, e-cash and more. The idea of a blind signature was first proposed by David Chaum [Cha83]. A blind signature is a way of obtaining a valid signature on a message without revealing anything about it. The user gets the blinded message signed by the signer and then publishes a valid signature on the original message in a way that no honest signer can tell when the message was provided to him for signing by seeing the message-signature pair.

In the past fifteen years, lattice-based cryptography has become a leading candidate for post-quantum security. The NIST has standardised one key encapsulation mechanism, CRYSTALS-Kyber [BDK+18], along with two digital signature schemes, namely Dilithium [DKL+18] and Falcon [PFH+17], that are based on lattice structures. In this paper, we propose an identity-based blind signature on the hardness of module-lattice problems. Designing an efficient blind signature protocol requires reducing the number of interaction rounds. The first lattice-based blind signature proposed in the paper [Rüc10] required four rounds of interaction. Later on, a few other lattice-based blind signatures [ABB20a, ABB20b, LSK+19, BCE+20, PHBS19, Rüc10] were also proposed. Some of these protocols were found to have erroneous security proofs, while many others were not round optimal. The paper [HKLN20] has proposed a lattice-based blind signature with appropriate security proofs; however, the resulting construction lacks practicality.

More recently, a line of research [dPK22, AKSY22, BLNS23] has adapted Fischlin's framework [Fis06] over a lattice setting, publishing a non-interactive zero-knowledge proof as the signature. This, in turn, gets rid of the unblinding step in the usual lattice-based versions and makes the protocol round optimal.

In this work, we are extending Fischlin's framework into an identity-based blind signature scheme, aiming to keep public keys as simple as possible and to eliminate certificates. The scheme we are proposing is round-optimal and the first of its kind which relies on the hardness of module lattice problems. The first ID-based blind signature based on pairings was introduced by Zhang and Kim [ZK02] followed by the papers [DHX21, SMKH10, RKP+10, GWWL12]. The first ID-based blind signature in the standard model over lattices was given in the paper [GHWX17]. Till date, there exists only a few works on ID-based blind signatures from lattice assumptions [GHW+17, SP20]. These constructions rely on classical lattice assumptions and do not provide practical parameter instantiations, limiting their efficiency and applicability in post-quantum settings. In this work, we present an identity-based blind signature scheme based on lattice trapdoor techniques and zero-knowledge proofs, combining G-trapdoor [MP12] constructions with zero-knowledge arguments to achieve blindness and unforgeability in the random oracle model.

1.1 Literature & our contribution

Our work is highly motivated and mostly in line with the constructions of the blind signatures in the papers [AKSY22, BLNS23]. We build on this by introducing a Trusted Authority (TA) who manages the generation and efficient distribution of keys. We use the G-trapdoor method for trapdoor generation, as described in the paper [MP12], to generate the keys in our scheme. We propose the concrete parameters for 128-bit security and our estimated signature size is around 330 KB.

Agrawal et al's work. In the paper [AKSY22], the authors have adapted Fischlin's framework [Fis06] to produce a blind signature on a message in a lattice setting. The blind signer initiates by generating a (PKE.pk, PKE.sk) and discarding PKE.sk. Then it samples a non-trapdoored matrix, often called CRS, $\mathbf{B} \in \mathbb{Z}_q^{n\times m}$, and $(\mathbf{C}, \mathbf{T_C}) \leftarrow \text{TrapdoorGen}(1^\lambda)$. The $\text{BSig.vk} = (\mathbf{C}, \mathbf{B}, \text{PKE.pk})$ and $\text{BSig.sk} = \mathbf{T_C}$. A user who wants to get a blind signature on a message μ, samples a random short vector $\mathbf{x}$ from $\mathrm{D}_{\mathcal{R}^m,\sigma_{\mathbf{x}}}$ with $\|\mathbf{x}\| \le \beta_{\mathbf{x}}$ and computes

$$\mathbf{t} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mu). \tag{1}$$

This $\mathbf{t}$ does not leak any information about $\mathbf{x}$ or the message μ because of "one time pad" style blinding mechanism. The blinded message $\mathbf{t}$ is then sent to the signer. The signer produces $\mathbf{y} \leftarrow \mathbf{C}_\sigma^{-1}(\mathbf{t})$ as a signature of $\mathbf{t}$, with the help of $\mathbf{T_C}$ and returns $\mathbf{y}$ to the user. A random string $r \in \{0,1\}^\lambda$ is sampled, and a ciphertext $\text{ct} := \text{PKE.enc}(\text{PKE.pk}, \mathbf{x}||\mathbf{y}; r)$ is computed using the randomness r. The user finally generates an NIZK proof

$$\pi : \left(\exists\, \mathbf{x}, \mathbf{y}, r \;\middle|\; \begin{array}{l} \mathbf{C}\mathbf{y} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mu) \wedge \\ \text{ct} = \text{PKE.enc}(\text{PKE.pk}, \mathbf{x} \parallel \mathbf{y}; r) \wedge \\ \|\mathbf{x}\| \le \beta_{\mathbf{x}} \wedge \|\mathbf{y}\| \le \beta_{\mathbf{y}} \wedge r \leftarrow \{0,1\}^\lambda \end{array} \right).$$

The user publishes $(\mu, (\pi, \text{ct}))$ as a signature of the message μ. The scheme is one-more-unforgeable based on the hardness assumption of the one-more-ISIS problem.

Lyubashevsky et al's work. In the paper [BLNS23], the authors modified the previous work in two ways. While blinding a message μ in Agrawal et al's framework, a user can compute $\mathrm{H}(\mu)$ and then choose $\mathbf{x}$ conditioned on $\mathrm{H}(\mu)$ so that it can influence the distribution of $\mathbf{t}$ in Equation (1) which further forces a signer to sign a non-uniform vector $\mathbf{t}$. To tackle this situation, Agrawal et al. introduced the new "one-more-ISIS" assumption, but due to lack of sufficient security reductions, the paper [BLNS23] modifies the blinding technique. A user who wants a blind signature on a message μ, samples a random short vector $\mathbf{x}$ from $\mathrm{D}_{\mathcal{R}^m,\sigma_{\mathbf{x}}}$ and computes a blinded message

$$\mathbf{t} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{G}(\mathbf{x}), \mu), \tag{2}$$

where G is a public hash function. This ensures that $\mathbf{t}$ is always uniform and thus the need for the one-more-ISIS problem is no longer relevant. An NIZK proof π_1 is associated to ensure the correctness of the construction of $\mathbf{t}$. The authors also introduced "encryption to the sky" of the witnesses $\mathbf{x}$ and μ and included the proof of the encryption in the NIZK. Encryption to the sky is a public key encryption scheme where the public and secret key pair is not generated using the key generation algorithm, instead the public key is sampled from a distribution of public keys, i.e., generated using some hash function. During the signature query of a user on the blinded message $\mathbf{t}$, $(\mathbf{t}, \mathrm{ct}, \pi_1)$ is sent to the blind signer. After successful verification of π_1, signer produces a Gaussian preimage $\mathbf{y}$ of the syndrome $\mathbf{t}$ using the trapdoor $\mathbf{T_C}$ and sends $\mathbf{y}$ to the user. An NIZK proof

$$\pi_2 : \left(\exists\, \mathbf{x}, \mathbf{y} \;\middle|\; \begin{array}{l} \mathbf{Cy} = \mathbf{Bx} + \mathrm{H}(\mathrm{G}(\mathbf{x}), \mu) \wedge \\ \|\mathbf{x}\| \leq \beta_{\mathbf{x}} \wedge \|\mathbf{y}\| \leq \beta_{\mathbf{y}} \end{array} \right)$$

together with $\rho = \mathrm{G}(\mathbf{x})$ i.e. (π_2, ρ) is generated as a signature for the message μ by the user. They have managed to reduce the signature size to 22KB by removing the ciphertext ct in the final signature .

In both works, a public-key encryption is needed for the security proof of "one-more-unforgeability". Without the ciphertext ct an extractor needs to extract the proof π or π_1, $Q_s + 1$ many times, which causes an exponential (in Q_s) security loss. In practical implementations, public-key encryption can be skipped.

Our work. We follow the blinding technique adopted by Lyubashevsky (see eq. 2) but replace the hash image $\mathrm{G}(\mathbf{x})$ with the Ajtai commitment $\mathrm{com}_{\mathbf{x}}$. Such modification enables us to move away from heuristic random-oracle assumptions toward the standard model and also the linear structure of $\mathrm{com}_{\mathbf{x}}$ makes it easier to be proved by lattice-based linear proof systems, overcoming the need of expensive circuit representations. Broadly, we replace the "programmable randomness" from hash functions with "structured pseudorandomness" based on lattice hardness. The user samples $\mathbf{x} \leftarrow \mathrm{D}_{\mathcal{R}^m, \sigma_{\mathbf{x}}}$ and computes

$$\mathbf{t} = \mathbf{Bx} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu) \tag{3}$$

where

$$\mathrm{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r} \text{ with } \|\mathbf{x}\| \leq \beta_{\mathbf{x}}, \|\mathbf{r}\| \leq \beta_{\mathbf{r}} \tag{4}$$

The blinding technique given in the Equation (3) ensures the randomness of $\mathbf{t}$. Moreover, given any uniform $\mathbf{t} \in \mathcal{R}_q^d$ and random commitment $\mathrm{com}_{\mathbf{x}}$, finding short vectors $\mathbf{x}$, $\mathbf{r}$ and a message μ reduces to solve Module-ISIS problem. The fact that the blinded message $\mathbf{t}$ is correctly computed is taken care of in the NIZK proof π_2 of the actual signature generation phase, as is done in the scheme [AKSY22]. This results in a reduction in the communication cost. The user then computes a ciphertext ct encrypting the secrets $\mathbf{x}, \mathbf{r}, \mu$ using the encryption randomness $\mathtt{rnd}$.

$$\mathrm{ct} = \mathrm{PKE.enc}(\mathrm{PKE.pk}, \mathbf{x} \parallel \mathbf{r} \parallel \mu; \mathtt{rnd}), \mathtt{rnd} \xleftarrow{\$} \{0,1\}^{\lambda} \tag{5}$$

and generates an NIZK proof π_1 for

$$\pi_1 : \left(\exists\, \mathbf{x}, \mathbf{r}, \mu, \mathtt{rnd} \;\middle|\; \begin{array}{l} \mathrm{ct} = \mathrm{PKE.enc}(\mathrm{PKE.pk}, \mathbf{x} \parallel \mathbf{r} \parallel \mu; \mathtt{rnd}) \wedge \\ \|\mathbf{x}\| \leq \beta_{\mathbf{x}} \wedge \|\mathbf{r}\| \leq \beta_{\mathbf{r}} \wedge \mathtt{rnd} \xleftarrow{\$} \{0,1\}^{\lambda} \end{array} \right), \tag{6}$$

which proves that the ciphertext is correctly generated. The ciphertext ct is needed in the security proof of our scheme, as it provides a witness-extraction facility. In practice, the ciphertext ct is generated using "encryption to the sky", as suggested in the paper [BLNS23]. For generating the public key for the encryption to the sky, we propose to use a hash function $\mathcal{H}_{\mathrm{PKE}} : \{0,1\}^{\star} \to \mathcal{K}_{\mathrm{PKE}}$ where $\mathcal{K}_{\mathrm{PKE}}$ is the public key space of the PKE. The user computes a

PKE.pk from $\mathcal{H}_{\text{PKE}}$ and sends ($\mathbf{t}$, π_1, ct) to the signer. Next, the signer samples a preimage for the syndrome $\mathbf{t}$ and sends $\mathbf{y}$ to the user. The commitment $\text{com}_{\mathbf{x}}$ is made public, hence a zero-knowledge proof is required to prove that the user knows a valid opening for the commitment. The final signature for a message μ set to be $(\text{com}_{\mathbf{x}}, \pi_2)$ where the NIZK proof π_2 is as follows,

$$\pi_2 : \left(\exists\, \mathbf{x}, \mathbf{y}, \mathbf{r} \;\middle|\; \begin{array}{l} \mathbf{C}\mathbf{y} = \mathbf{B}\mathbf{x} + \text{H}(\text{com}_{\mathbf{x}}, \mu) \wedge \\ \text{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r} \wedge \\ \|\mathbf{x}\| \leq \beta_{\mathbf{x}} \wedge \|\mathbf{y}\| \leq \beta_{\mathbf{y}} \wedge \|\mathbf{r}\| \leq \beta_{\mathbf{r}} \end{array} \right). \tag{7}$$

ID**-based Blind Signature Framework.** In this section, we propose an ID-based blind signature scheme on module lattices based on the above considerations. We begin with a trusted authority (TA) that initiates a list $\mathcal{ID}$ of ID's and assigns them to the blind signers. The TA sets up the entire protocol by constructing a master public key (mpk) $\mathbf{A} := [\bar{\mathbf{A}}|\mathbf{G} - \bar{\mathbf{A}}\mathbf{R}] \in \mathcal{R}_q^{d\times m}$, $m = 2d + dk$, together with a $\mathbf{G}$-trapdoor $\mathbf{R} \in \mathcal{R}^{2d\times dk}$ where $\mathbf{G} = \mathbf{I}_n \otimes \mathbf{g}^\top \in \mathcal{R}_q^{d\times dk}$, $\mathbf{g}^\top = \left(1, 2, 2^2, \ldots, 2^{k-1}\right) \in \mathcal{R}_q^k$ and $\bar{\mathbf{A}} \xleftarrow{\$} \mathcal{R}_q^{d\times 2d}$ using the algorithm TrapGen(1^λ) and sets the trapdoor $\mathbf{R}$ as a master secret key (msk). The msk will eventually be used to generate signing keys for the IDs. For a given ID $\in \mathcal{ID}$, TA generates a matrix $\mathcal{H}(\text{ID}) \in \mathbb{Z}_q^{d\times dk}$ and constructs a $\mathbf{G}$-trapdoor $\mathbf{R}_{\text{ID}} \in \mathcal{R}^{m\times dk}$ with the help of msk and the algorithm, DelTrap$(\mathbf{A}_{\text{ID}}, \mathbf{G}, s)$ where $\mathbf{A}_{\text{ID}} = [\mathbf{A}|\mathcal{H}(\text{ID})] \in \mathcal{R}_q^{d\times(m+dk)}$, s is a sampling parameter and $\mathcal{H}(\cdot)$ is a cryptographic hash function. In the ID-based setting, TA publishes a hash function $\mathcal{H}_{\text{PKE}} : \{0,1\}^\star \longrightarrow \mathcal{K}_{\text{PKE}}$, which an user uses to generate PKE.pk $:= \mathcal{H}_{\text{PKE}}(\text{ID})$. Moreover, $\mathcal{H}_{\text{PKE}}$ is modelled as a random oracle in the security proof of the IBS's unforgeability. The public parameter pp is set to be $(\lambda, \mathbf{G}, \mathbf{A}, \mathbf{B}, \mathbf{A}_1, \mathbf{A}_2, \mathcal{H}, \text{H}, \mathcal{H}_{\text{PKE}})$, where λ is a security parameter and $\mathbf{B} \in \mathcal{R}_q^{d\times m}$, $\mathbf{A}_1 \in \mathcal{R}_q^{d\times m}$ and $\mathbf{A}_2 \in \mathcal{R}_q^{d\times 1}$ are non-trapdoored uniform matrices sampled by TA. The signing and verification keys are set to be IDBSig.sk $:= \mathbf{R}_{\text{ID}} \in \mathcal{R}^{m\times dk}$ and IDBSig.vk := ID respectively. Now, following the blind signature framework described above, the user $\mathcal{U}$ sends ($\mathbf{t}$, π_1, ct) to the signer $\mathcal{S}$ with the corresponding ID. Upon receiving $\mathbf{t}$, signer uses the delegated trapdoor $\mathbf{R}_{\text{ID}}$ to sample a short $\mathbf{y}$ with $\|\mathbf{y}\| \leq \beta_y$ such that $\mathbf{A}_{\text{ID}}\mathbf{y} = \mathbf{t} \pmod q$. The signer sends $\mathbf{y}$ to the user.

Upon receiving $\mathbf{y}$, the user verifies that $\|\mathbf{y}\| \leq \beta_y$ and $\mathbf{A}_{\text{ID}}\mathbf{y} = \mathbf{t} \pmod q$; if verification fails, the user aborts the rest of the protocol. The user publishes $(\mu, (\text{com}_{\mathbf{x}}, \pi_2))$ as the final message-signature pair where, given IDBSig.vk =ID, μ, and pp, the NIZK proof π_2 is,

$$\pi_2 : \left(\exists\, \mathbf{x}, \mathbf{y}, \mathbf{r} \;\middle|\; \begin{array}{l} \mathbf{A}_{\text{ID}}\mathbf{y} = \mathbf{B}\mathbf{x} + \text{H}(\text{com}_{\mathbf{x}}, \mu, \text{ID}) \wedge \\ \text{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r} \wedge \\ \|\mathbf{x}\| \leq \beta_{\mathbf{x}} \wedge \|\mathbf{y}\| \leq \beta_{\mathbf{y}} \wedge \|\mathbf{r}\| \leq \beta_{\mathbf{r}} \end{array} \right)$$

The paper [AKSY22] uses Falcon (0.6 KB) as the basic signature scheme and produces a signature of size 45.19 KB. Instead, we build our signature scheme from the G-trapdoor-based GPV signature [BEP+21] where the size of the vector $\mathbf{y}$ is around 98.8 KB. This highly influences the final signature size of our scheme, which is around 330.66 KB. Additionally, our scheme introduces a TA for better key management, which delegates the signing key to the signer, further causing an increase in signature size. To instantiate the whole scheme over a module lattice, we use the trapdoor generation, Gaussian preimage sampling, and GPV-signature from the papers [GM17, BEP+21]. The hash function $\mathcal{H}_{\text{PKE}}$ generates PKE.pk of the PKE [BDK+18]. The proof π_1 can also be instantiated using the linear proof system from the paper [LNP22] whereby we completely avoid the overhead of circuit based proof systems. However, since this does not contribute to the signature size, we leave the concrete instantiation of π_1 as future work. We provide a detailed study of the scheme parameters along with the NIZK proof π_2 [LNP22] in Section 4.2 and a Python script, adapted from the paper [AKSY22], for verification of our claimed parameters. Our scheme is secure under the assumption that MLWE and MSIS are hard.

2 Preliminaries

2.1 Notation

We denote the rings of integers and integers modulo q by $\mathbb{Z}$ and $\mathbb{Z}_q$ respectively. We use $\mathbb{R}$ and $\mathbb{C}$ for the fields of real and complex numbers, respectively. The elements of these sets are denoted by standard lowercase letters, (columns) vectors by bold lowercase letters, and matrices by bold uppercase letters. The norm $\|\cdot\|$ denotes the Euclidean norm, and the norm of a vector over $\mathbb{Z}_q$ is the norm of the corresponding vector over $\mathbb{Z}$ with entries in $\{-\lfloor q/2 \rfloor, \ldots, \lfloor q/2 \rfloor\}$, the norm of a polynomial $a = \sum_{i=1}^{n-1} a_i X^i$ is the norm of the vector $(a_0, a_1, \ldots, a_{n-1})$. The norm of a matrix is defined to be the maximum norm of its column vectors. We use $x \leftarrow \mathrm{D}$ to mean that x is sampled from a discrete distribution D. If the distribution is uniform with finite support S, then we write $x \stackrel{\$}{\leftarrow} S$. We denote the uniform distribution on S by $\mathcal{U}(S)$. For $a \in \mathbb{Z}_q$, the binary representation of a, denoted by $[a]_2^k$, is thought of as a k dimensional vector $\mathbf{a} = (a_0, a_1, \ldots, a_{k-2}, a_{k-1})$ where $k = \lceil \log_2 q \rceil$. We use $\langle \cdot, \cdot \rangle$ to denote interactive execution of two algorithms. By $(a, b) \leftarrow \langle \mathcal{X}(x), \mathcal{Y}(y) \rangle$, we mean the joint execution of $\mathcal{X}$ and $\mathcal{Y}$ where the algorithm $\mathcal{X}$ has private input x and the algorithm $\mathcal{Y}$ has private input y, and $\mathcal{X}$ receives private output a while $\mathcal{Y}$ receives private output b.

2.2 ID-Based Blind Signatures

Definition 1. *(Blind Signature) A blind signature scheme* BS *consists of* PPT *algorithms* (Setup, Verification) *along with interactive* PPT *algorithms* $\mathcal{S}$ *and* $\mathcal{U}$*, where*

1. Setup(n)*: On input a security parameter n, it generates a keypair* (BSig.sk, BSig.vk).
2. *The joint execution of* $\langle \mathcal{S}(\text{BSig.sk}), \mathcal{U}(\text{BSig.vk}, \mu) \rangle$*, where* $\mu \in \{0,1\}^\star$ *is a message, generates a signature* σ_μ *for the user and no output for the signer; we denote it by* $(\perp, \sigma_\mu) \leftarrow \langle \mathcal{S}(\text{BSig.sk}), \mathcal{U}(\text{BSig.vk}, \mu) \rangle$.
3. Verification*: On input* $(\text{BSig.vk}, \mu, \sigma_\mu)$*, it outputs a bit* 1 *for "accept" or* 0 *for "reject".*

An ID-based blind signature scheme is an extension of the concept of a blind signature which simplifies the requirement of public key management step by introducing a trusted authority (TA) that performs the Setup step to generate the public parameters (pp) and the master key pair (mpk, msk). The signer needs to perform an interactive step KeyExtract with the TA to extract its signing key $\mathcal{S}_{\text{ID}}$ corresponding to its identity ID, issued beforehand by the TA. The verification key is simply the public identity ID. Thus, an ID-based blind signature IBBS is a collection of four PPT algorithms Setup, $\langle \mathcal{S}(\text{ID}), \text{TA}(mpk, msk) \rangle$, $\langle \mathcal{S}(\text{IDBSig.sk}), \mathcal{U}(\text{IDBSig.vk}, \mu) \rangle$, and Vrfy. It must satisfy two security properties: blindness and selective-ID one-more unforgeability [GHW+17, ZHTY10].

Definition 2. *(Honest Signer Blindness) The honest signer blindness property enables the malicious signer* $\mathcal{S}^\star$ *to sign two messages* μ_0, μ_1 *of its choice in such a way that* $\mathcal{S}^\star$ *will never be able to decide which message has been signed first in two executions with an honest user* $\mathcal{U}$.

Definition 3. *(sID - One More Unforgeability) Let* $\mathcal{U}^\star$ *be a* PPT *malicious algorithm (user) participating in the following game with the challenger.*

1. $\mathcal{U}^\star$ *chooses an identity, say* $\text{ID}^\star \in \mathcal{ID}$.
2. *The TA generates the public parameters* (pp) *and the master key pair* (mpk, msk).
3. $\mathcal{U}^\star$ *can perform two types of queries. Key Extraction Query (to the TA):* $\mathcal{U}^\star$ *can extract the signing key* $\mathcal{S}_{\text{ID}}$ *corresponding to any* $\text{ID} \in \mathcal{ID} \setminus \{\text{ID}^\star\}$*; and Signing Query (to* $\mathcal{S}$*):* $\mathcal{U}^\star$ *can make signing queries for blinded messages of its choice under the* ID' *in* $\mathcal{ID} \setminus \{\text{ID}^\star\}$.

4. *After polynomially many key extraction and signature queries, the algorithm $\mathcal{U}^\star$ outputs the tuple $(\mu^\star, \sigma_{\mu^\star}, \mathrm{ID}^\star)$, where $\sigma_{\mu^\star}$ is the signature on the message $\mu^\star$ which was never queried before) under $\mathrm{ID}^\star$, chosen by $\mathcal{U}^\star$ before seeing the public parameters.*

The Algorithm $\mathcal{U}^\star$ succeeds iff $\mathrm{Vrfy}(\mathrm{ID}^\star, \mu^\star, \sigma_{\mu^\star}) = 1$, i.e. the $\mathcal{U}^\star$ outputs a valid signature of a message under its chosen identity without knowing the signing key. We say that the IBBS *satisfies selective-ID one-more unforgeable if $\mathcal{U}^\star$ succeeds with negligible probability.*

2.3 Commitment Scheme

Definition 4. *(Commitment Scheme) A commitment scheme consists of probabilistic polynomial time algorithms* (Setup, Gen, Commit, Open) *such that:*

1. $\mathrm{cpp} \leftarrow \mathrm{Setup}(1^k)$ *: The setup algorithm outputs a public parameter* cpp *and defines sets $S_{\mathrm{ck}}, S_{\mathrm{msg}}, S_r$ and S_{com} and the distribution $D(S_r)$ from which the randomness is sampled.*
2. $\mathrm{ck} \leftarrow \mathrm{Gen}(\mathrm{cpp})$ *: The key generation algorithm that samples a commitment key from S_{ck}. We often omit mentioning the public key ck when it is clear from the context.*
3. $\mathrm{com} \leftarrow \mathrm{Commit}_{\mathrm{ck}}(\mathrm{msg}; r)$ *: The committing algorithm that takes a message* $\mathrm{msg} \in S_{\mathrm{msg}}$ *and randomness $r \in S_r$ as input and outputs* $\mathrm{com} \in S_{\mathrm{com}}$.
4. $(\mathrm{msg}, r) \leftarrow \mathrm{Open}_{\mathrm{ck}}(\mathrm{com})$ *: The opening algorithm takes input the commitment and open the* msg *and randomness r.*

Upon receiving the commitment com *and its opening* (msg, r)*, one can check the validity of the commitment. The receiver outputs a bit b; $b := 1$ if* com *is a valid commitment, otherwise $b := 0$.*

The zero-knowledge opening proof of a commitment scheme is a zero-knowledge protocol that proves knowledge of the committed secret values given the public commitment.

Security. Any secure commitment scheme should have two properties. One is hiding, which ensures that the commitment reveals no information about the secret. Another is binding, which ensures that the committer cannot change the committed secret after the commitment is published. These can be formally defined as,

1. **Hiding.** The commitment $\mathrm{com} = \mathrm{Commit}_{\mathrm{ck}}(\mathrm{msg}; r)$ must not reveal any information about the msg, thus,

$$|\Pr[\mathrm{com} = \mathrm{Commit}_{\mathrm{ck}}(\mathrm{msg}; r)] - \Pr[\mathrm{com} \xleftarrow{\$} S_{\mathrm{com}}]| < \epsilon$$

2. **Binding.** A commitment scheme is said to be (t, ϵ)-secure if $\forall \mathrm{ck} \in S_{\mathrm{ck}}$, for all Open algorithms that run in time t and output $\mathrm{Open}_{\mathrm{ck}}(\mathrm{com}) = (\mathrm{msg}_1, \mathrm{msg}_2, r_1, r_2)$, we have

$$\Pr[\mathrm{msg}_1 \neq \mathrm{msg}_2, \mathrm{Commit}_{\mathrm{ck}}(\mathrm{msg}_1; r_1) = \mathrm{Commit}_{\mathrm{ck}}(\mathrm{msg}_2; r_2)] < \epsilon$$

Ajtai commitment scheme [Ajt96]. We can commit to a short secret $\mathbf{s}_1$ using short randomness $\mathbf{s}_2$ as follows:

$$\mathbf{A}_1\mathbf{s}_1 + \mathbf{A}_2\mathbf{s}_2 = \mathbf{t} \pmod q \tag{8}$$

Here, $\mathbf{t} = \mathrm{Commit}_{\mathbf{A}_1,\mathbf{A}_2}(\mathbf{s}_1; \mathbf{s}_2)$ is the Ajtai commitment to $\mathbf{s}_1$. An opening of this commitment is $(\mathbf{s}_1, \mathbf{s}_2)$ and Lemma 4 and SIS hardness provide the necessary hiding and binding properties.

2.4 Non-Interactive Zero Knowledge

Any Σ-protocol for an NP relation $\mathrm{R} \subseteq \mathcal{W} \times \mathcal{S}$ is a pair of algorithms (P, V) where P proves knowledge of some witness $w \in \mathcal{W}$ to a statement $s \in \mathcal{S}$ to the V such that $(w, s) \in \mathrm{R}$. Any such protocol can be transformed into a non-interactive protocol using the Fiat-Shamir transform.

Definition 5. *(Non Interactive Zero Knowledge) An non-interactive zero-knowledge proof* π *for an NP relation* R *consists of two* PPT *algorithms* (P, V) *with the following properties:*

1. $\mathrm{P}(w, s) \to \pi$ *: The algorithm* P *takes input a witness* w *for a statement* s *such that* $(w, s) \in \mathrm{R}$ *and outputs a proof* π*.*

2. $\mathrm{V}(s, \pi) \to \mathrm{accept/reject}$ *: The verifier* V *takes input a statement* x *and a proof* π *for that statement and outputs* accept/reject*.*

The proof π should satisfy the following properties.

1. **Completeness:** This means that an honest prover can convince an honest verifier that the statement is true i.e., for any $(w, s) \in \mathrm{R}$, we have
$$\Pr[\pi \leftarrow \mathrm{P}(w, s) : \mathrm{V}(s, \pi) = 1] \geq 1 - \mathrm{negl}.$$

2. **Soundness:** This means that verifying a proof of a false statement can never succeed i.e., for any $s \in \mathcal{S}$ such that $(w, s) \notin \mathrm{R}$ and any poly-time dishonest prover $\mathrm{P}^\star$, we have
$$\Pr[\pi \leftarrow \mathrm{P}^\star(s) : \mathrm{V}(s, \pi) = 1] \leq \mathrm{negl}$$

3. **Honest Verifier Zero Knowledge:** This property should hold so that an honest verifier learns only that the statement is true and has zero knowledge about the secret. If there is a PPT simulator Sim such that, for all statements x such that there exists w with $(w, s) \in \mathrm{R}$, then for any poly-time adversary $\mathcal{A}$, we have :
$$\left| \begin{array}{l} \Pr[1 \leftarrow \mathcal{A}(s, \pi) : \pi \leftarrow \mathrm{P}(w, s))] - \\ \qquad \Pr[1 \leftarrow \mathcal{A}((s, \pi) : \pi \leftarrow \mathrm{Sim}(s))] \end{array} \right| \leq \mathrm{negl}$$

4. ***Knowledge Sound:*** For any polynomial-time prover $\mathrm{P}^\star$, there exists an extractor $\mathcal{E}$ with expected run-time polynomial in λ and the run-time of $\mathrm{P}^\star$,
$$\Pr\left[\begin{array}{l} \mathrm{crs} \leftarrow \mathrm{Gen}(1^\lambda), \\ (x, s) \leftarrow \mathcal{A}(\mathrm{crs}), \\ \pi^\star \leftarrow P^\star(\mathrm{crs}, x, s), \\ b \leftarrow V(\mathrm{crs}, x, \pi^\star), \\ w \leftarrow \mathcal{E}^{P^\star(\mathrm{crs}, x, s)}(\mathrm{crs}, x, \pi^\star, b) \end{array} \middle| \; (x, w) \notin R \wedge b = \mathrm{accept} \right] \leq \mathrm{negl}.$$

2.5 Linear Algebra

A symmetric matrix $\mathbf{M} \in \mathbb{R}^{m \times m}$ is called to be positive definite (resp. positive semidefinite) if for all nonzero $\mathbf{v} \in \mathbb{R}^m$ we have $\mathbf{v}^\top \mathbf{M} \mathbf{v} > 0$ (resp. $\mathbf{v}^\top \mathbf{M} \mathbf{v} \geq 0$), in which case we denote $\mathbf{M} > 0$ (resp. $\mathbf{M} \geq 0$). If $\mathbf{M}$ and $\mathbf{N}$ be two real symmetric matrices then we write $\mathbf{M} > \mathbf{N}$ (resp. $\mathbf{M} \geq \mathbf{N}$) if $\mathbf{M} - \mathbf{N} > 0$ (resp. $\mathbf{M} - \mathbf{N} \geq 0$). If $\mathbf{N} = s\mathbf{I}$, we write $\mathbf{M} \geq s$ for any non-negative real s . For any real matrix $\mathbf{B}^{m \times n}$, $\Sigma = \mathbf{B}\mathbf{B}^\top$ is always symmetric and positive semidefinite because $\mathbf{v}^\top \Sigma \mathbf{v} = \langle \mathbf{B}^\top \mathbf{v}, \mathbf{B}^\top \mathbf{v} \rangle = \|\mathbf{B}^\top \mathbf{v}\|^2 \geq 0$ for any nonzero $\mathbf{v} \in \mathbb{R}^m$. The inequality is strict if and only if $\mathbf{B}$ is invertible. We say that $\mathbf{B}$ is a square root of Σ and denote $\mathbf{B} = \sqrt{\Sigma}$. Every positive definite (resp. positive semidefinite) matrix Σ has a square root because every real symmetric matrix is orthogonally diagonalizable. So if $\Sigma > 0$ (resp. $\Sigma \geq 0$), and being a real symmetric matrix $\Sigma = \mathbf{U}^\top \mathbf{D} \mathbf{U}$ for some orthogonal matrix and $\mathbf{D} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_m)$ is a diagonal matrix with positive eigenvalues (resp. non-negative eigenvalues) $(\lambda_i)_{i=1}^m$. Now we write $\mathbf{D} = \sqrt{\mathbf{D}}\sqrt{\mathbf{D}}^\top$ where $\sqrt{\mathbf{D}} = \mathrm{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_m})$ and we have $\sqrt{\Sigma} = \left(\sqrt{\mathbf{D}}^\top \mathbf{U}\right)^\top$.

2.6 Lattices and Gaussians

Given any linearly independent subset $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\}$ of $\mathbb{R}^m$, $k \leq m$. A *lattice* Λ of rank k is defined as $\Lambda(\mathbf{B}) := \left\{\sum_{i=1}^{k} z_i \mathbf{b}_i \mid z_i \in \mathbb{Z}\right\}$. The *dual lattice* of Λ , denoted by $\Lambda^\star$, is another lattice $\{\mathbf{v} \in \text{span}(\mathbf{B}) | \langle \mathbf{v}, \mathbf{w}\rangle \in \mathbb{Z},\ \forall \mathbf{w} \in \Lambda\}$. If we take $(k = m)$, then Λ is called a full rank lattice. It is easy to see that for a full rank lattice Λ , the $\Lambda^\star$ has a basis $\mathbf{B}^{-\top}$. For $\mathbf{A} \in \mathbb{Z}^{n\times m}$ and $\mathbf{u} \in \mathbb{Z}_q^n$, we define the following m-dimensional q-ary lattice and its coset $\Lambda_q^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m : \mathbf{A}\mathbf{x} = \mathbf{0} \pmod q\}$ and $\Lambda_q^{\mathbf{u}}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m : \mathbf{A}\mathbf{x} = \mathbf{u} \pmod q\}$.
The m-dimensional Gaussian function $\rho : \mathbb{R}^m \to (0, 1]$ is defined by $\rho(\mathbf{v}-\mathbf{c}) := \exp\left(-\pi\|\mathbf{v} - \mathbf{c}\|^2\right)$ for any fixed $\mathbf{c} \in \mathbb{R}^m$. Applying an invertible linear transformation $\mathbf{B}$ to ρ yields:

$$\rho_{\mathbf{B},\mathbf{c}}(\mathbf{v}) := \exp\left(\mathbf{B}^{-1}(\mathbf{v} - \mathbf{c})\right) = \exp\left(-\pi(\mathbf{v} - \mathbf{c})^\top \Sigma^{-1}(\mathbf{v} - \mathbf{c})\right)$$

with $\Sigma = \mathbf{B}\mathbf{B}^\top > 0$. Normalizing the function $\rho_{\mathbf{B},\mathbf{c}}(\mathbf{v})$ by the measure of $\rho_{\mathbf{B},\mathbf{c}}$ over the span$(\mathbf{B})$ provide us the *continuous Gaussian distribution* with covariance $\frac{\Sigma}{2\pi}$, having centre at $\mathbf{c}$, denoted by $\mathrm{D}_{\sqrt{\Sigma},\mathbf{c}}$. When $\mathbf{c} = \mathbf{0}$, we simply denote $\mathrm{D}_{\sqrt{\Sigma}}$. Let A be any discrete subset of $\mathbb{R}^m$, then $\rho_{\sqrt{\Sigma}}(A) = \Sigma_{a\in A}\ \rho_{\sqrt{\Sigma}}(a)$. The discrete Gaussian over a lattice Λ, denoted by $\mathrm{D}_{\Lambda,\Sigma,\mathbf{c}}$, is defined by restricting the support of the distribution to Λ. The discrete Gaussian distribution of center $\mathbf{c} \in \mathbb{R}^n$ and parameter $\sigma > 0$ over a full-rank lattice $\Lambda \subset \mathbb{Z}^n$ is denoted $\mathrm{D}_{\Lambda,\sigma,\mathbf{c}}$. It is the probability distribution over Λ such that each $\mathbf{x} \in \Lambda$ is assigned a probability proportional to

$$\rho_{\sigma,\mathbf{c}}(\mathbf{x}) = \exp\left(-\pi\frac{\|\mathbf{x} - \mathbf{c}\|^2}{\sigma^2}\right).$$

For any lattice Λ and any positive ϵ, the *smoothing parameter* $\eta_\epsilon(\Lambda)$ [MR07] is the smallest $s > 0$ such that $\rho(s \cdot \Lambda^\star) \leq 1 + \epsilon$.

Tail-cut

A one-dimensional discrete Gaussian with a tail-cut, t, is a discrete Gaussian $\mathrm{D}_{\mathbb{Z},c,s}$ restricted to a support of $\mathbb{Z} \cap [c - t \cdot s, c + t \cdot s]$. To tail-cut less than $2^{-\lambda}$ of a one-dimensional Gaussian, we use the fact that

$$\Pr_{x\leftarrow D_{\mathbb{Z},\sigma}}\left[\,|x| > t\sigma\,\right] \leq \operatorname{erfc}\left(\frac{t}{\sqrt{2}}\right),$$

where

$$\operatorname{erfc}(x) = 1 - \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt.$$

For example, $t = 15$ for $\lambda = 192$. Then, a vector $\mathbf{x}$ sampled in $D_{\mathbb{Z}^m,\sigma}$ would have small norm $\|\mathbf{x}\| \leq t\sigma\sqrt{m}$ with overwhelming probability.

Module Lattice

Let F be a number field of degree n i.e., $F := \mathbb{Q}[X]/\langle f(X)\rangle$ for a monic irreducible polynomial f over $\mathbb{Q}$ of degree n and $\sigma_i : F \to \mathbb{C}$ be monomorphisms for $i, 1 \leq i \leq n$. We enumerate the monomorphisms in such a way that $\sigma_1, \sigma_2, \ldots, \sigma_r$ are real (i.e. $\sigma_i(F) \subseteq \mathbb{R}$) and $\sigma_{r+1}, \sigma_{r+2}, \ldots, \sigma_{r+s}$ are complex (i.e. $\sigma_i(F) \subseteq \mathbb{C}$) and $\sigma_{r+s+1} = \overline{\sigma_{r+1}}, \sigma_{r+s+2} = \overline{\sigma_{r+2}}, \ldots, \sigma_n = \overline{\sigma_{r+s}}$. Note that $r + 2s = n$. We define $\sigma : F \to \mathbb{C}^n$ by $\sigma(\alpha) := (\sigma_i(\alpha))_{i=1}^n \forall\alpha \in F$ and call σ as *canonical embedding*. Let $\mathcal{R}$ be the corresponding number ring. A *module lattice* over $\mathcal{R}$ is the set of all $\mathcal{R}$-linear combinations of finitely many generating vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m \in F^\ell$,

$$\mathcal{M} = \{x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_m\mathbf{v}_m : x_i \in \mathcal{R}\}.$$

By the canonical embedding, we can view the module lattice $\mathcal{M}$ as an ℓn-dimensional integer lattice.

For any $a \in F$, $a = \sum_{i=0}^{\varphi(n)} a_i X^i \mapsto (a_0, a_1, \ldots, a_{\varphi(n)})$. It is called *coefficient embedding* under the basis $\left(1, X, X^2, \ldots, X^{\varphi(n)-1}\right)$. Let us define a linear map for a fixed element $a \in F$ and $\forall b \in F$, by $f(b) = a \cdot b$. We can represent a field element as a matrix in $\mathbb{Q}^{\varphi(n) \times \varphi(n)}$, which is the matrix representation of the map f.

In this paper, we use $f(X) = X^n + 1, n = 2^k, k \in \mathbb{Z}^+$. So $F = \mathbb{Q}[X]/\langle X^n + 1\rangle$, $\mathcal{R} = \mathbb{Z}[X]/\langle X^n + 1\rangle$ and $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle X^n + 1\rangle$ where q is a prime. An interesting fact is that when n is a power of two, the linear transformation between the coefficient embedding and the canonical embedding is a scaled isometry.

For any $a = \sum_{i=0}^{n-1} a_i X^i \in \mathbb{Q}[X]/\langle X^n + 1\rangle$, the matrix representation of the map f is given by the anti-circulant matrix

$$\mathrm{Ant}(a) = \begin{bmatrix} a_0 & -a_{n-1} & \cdots & -a_1 \\ a_1 & a_0 & \cdots & -a_{n-2} \\ \vdots & \vdots & \ddots & -a_{n-1} \\ a_{n-1} & a_{n-2} & \cdots & a_0 \end{bmatrix}.$$

For any element $a \in \mathcal{P} = \mathbb{R}[X]/\langle X^n + 1\rangle$, $a \mapsto \mathrm{Ant}(a)$ is an injective ring homomorphism. Sampling a vector consisting of m polynomials with a covariance matrix $\Sigma \in \mathcal{P}^{m \times m}$ is then equivalent to sampling a vector of mn length with a covariance matrix of order $mn \times mn$ over $\mathbb{R}$.

2.7 Min-Entropy

Let $\mathcal{C}$ be a countable set and $\mathcal{P}$ be a discrete probability distribution over $\mathcal{C}$. We define the collision probability of $\mathrm{Coll}(\mathcal{P}) = \Pr[y = y' : y, y' \leftarrow \mathrm{P}]$. By $\mathbb{E}(\mathcal{P})$, we denote the expected value of $\mathcal{P}$. The collision probability equals $\mathrm{Coll}(\mathcal{P}) = \Sigma_{y \in \mathcal{C}} \mathcal{P}(y)^2$ and the expected value equals $\mathbb{E}(\mathcal{P}) = \Sigma_{y \in \mathcal{C}}\, y \cdot \mathcal{P}(y)$. The min-entropy of $\mathcal{P}$ is defined as $\mathrm{H}_\infty(\mathcal{P}) = -\log_2\left(\max_{y \in \mathcal{S}} \mathcal{P}(y)\right)$. It is true that $\mathrm{Coll}(\mathcal{P}) \leq \Sigma_{x \in \mathcal{S}} \mathcal{P}(x) \cdot \max_{y \in \mathcal{S}} \mathcal{P}(y) = \max_{y \in \mathcal{S}} \mathcal{P}(y)$, which implies $\mathrm{Coll}(\mathcal{P}) \leq 2^{-\mathrm{H}_\infty(\mathcal{P})}$.

2.8 Hardness Assumptions

Definition 6 (**Module-ISIS**$_{n,d,m,q,\beta}$)**.** *Given a uniformly random $\boldsymbol{A} \in \mathcal{R}_q^{d \times m}$ and $\boldsymbol{u} \in \mathcal{R}_q^d$, find a vector $\boldsymbol{x} \in \mathcal{R}^m$ such that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{u} \mod q$, and $0 < \|\boldsymbol{x}\| \leq \beta$.*

Definition 7 (**Decision Module-LWE**$_{n,d,q,\sigma}$)**.** *Let $\boldsymbol{A} \xleftarrow{\$} \mathcal{R}_q^{d \times m}$ be uniformly random, and let*

$$\boldsymbol{b} = \boldsymbol{A}^\top \boldsymbol{s} + \boldsymbol{e} \bmod q,$$

where $\boldsymbol{s} \xleftarrow{\$} \mathcal{U}(\mathcal{R}_q^d)$ and $\boldsymbol{e} \xleftarrow{\$} D_{\mathcal{R}^m,\sigma}$. The Decision Module-LWE$_{n,d,q,\sigma}$ *problem is to distinguish the distribution of $(\boldsymbol{A}, \boldsymbol{b})$ from the uniform distribution over $\mathcal{R}_q^{m \times d} \times \mathcal{R}_q^m$.*

2.9 Useful Lemmas

Lemma 1 ([GPV07])**.** *For any n-dimensional lattice Λ, $\mathbf{c} \in \mathrm{span}(\Lambda)$, $\epsilon \in (0,1)$, and $s \geq \eta_\epsilon(\Lambda)$,*

$$\Pr_{\mathbf{x} \sim D_{\Lambda,s,\mathbf{c}}}\left[\|\mathbf{x} - \mathbf{c}\| > s\sqrt{n}\right] \leq \frac{1+\varepsilon}{1-\varepsilon} \cdot 2^{-n}.$$

Lemma 2 ([PR06])**.** *For any n-dimensional lattice Λ, center $\mathbf{c} \in \mathbb{R}^n$, positive $\varepsilon > 0$, and $s \geq 2\eta_\varepsilon(\Lambda)$, and for every $\mathbf{x} \in \Lambda$, we have*

$$D_{\Lambda,s,\mathbf{c}}(\mathbf{x}) \leq \frac{1+\varepsilon}{1-\varepsilon} \cdot 2^{-n}.$$

In particular, for $\varepsilon < \frac{1}{3}$, the min-entropy of $D_{\Lambda,s,\mathbf{c}}$ is at least $n-1$.

Lemma 3 ([GPV07]). *Assume the columns of $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ generate $\mathbb{Z}_q^n$, and let $\varepsilon \in (0, \frac{1}{2})$ and $s \geq \eta_\varepsilon(\Lambda^\perp(\mathbf{A}))$. Then for $\mathbf{e} \sim D_{\mathbb{Z}^m,s}$, the distribution of the syndrome $\mathbf{u} = \mathbf{A}\mathbf{e} \bmod q$ is within statistical distance 2ε of uniform over $\mathbb{Z}_q^n$.*

Lemma 4 (Left Over Hash Lemma). *Let $\mathbb{H} = \{h : \mathcal{X} \to \mathcal{Y}\}$ be a 2-universal hash function family. Then for any $\epsilon > 0$ and any random variable $X \in \mathcal{X}$ such that $\log|\mathcal{Y}| \leq H_\infty(X) - 2\log(1/\epsilon)$, the distributions*

$$(h, h(X)) \ \text{ and } \ (h, \mathcal{U}(\mathcal{Y}))$$

are within statistical distance ϵ.

Moreover, the family $\{\boldsymbol{A} \in \mathbb{Z}_p^{n\times m} : \boldsymbol{r} \mapsto \boldsymbol{A}\boldsymbol{r}\}$ is 2-universal for any prime p.

Lemma 5 ([GPV07]). *Let $\Lambda \subset \mathbb{R}^n$ be a lattice with basis $\boldsymbol{B}$ and $\epsilon > 0$. We have*

$$\eta_\epsilon(\Lambda) \leq \|\tilde{\boldsymbol{B}}\| \cdot \sqrt{\ln\left(2n(1+1/\epsilon)\right)/\pi}.$$

Particularly, for any $\omega(\sqrt{\log n})$ function, there is a negligible function $\epsilon(n)$ for which $\eta_\epsilon(\Lambda) \leq \|\tilde{\boldsymbol{B}}\| \cdot \omega(\sqrt{\log n})$.

Theorem 1 ([GPV07]). *Given a full-rank lattice basis $\mathbf{B} \in \mathbb{R}^{n\times n}$, a parameter $s \geq \|\tilde{\mathbf{B}}\|\, \omega(\sqrt{\log n})$, and a center $\mathbf{c} \in \mathbb{R}^n$, there is an $O(n^2)$-time, with an $O(n^3)$-time preprocessing, probabilistic algorithm whose output is statistically close to $D_{\mathcal{ID}(\mathbf{B}),\,s,\,\mathbf{c}}$.*

3 Gadget Trapdoor and Gaussian Preimage Sampling over Module Lattices

The *gadget vector* and *gadget matrix* are defined as:

$$\mathbf{g}^\top = \left(1, 2, 2^2, \ldots, 2^{k-1}\right) \in \mathcal{R}_q^{1\times k}$$

and

$$\mathbf{G} = \mathbf{I}_d \otimes \mathbf{g}^\top = \begin{pmatrix} \mathbf{g}^\top & & & \\ & \mathbf{g}^\top & & \\ & & \ddots & \\ & & & \mathbf{g}^\top \end{pmatrix} \in \mathcal{R}_q^{d\times dk}$$

where $k = \lceil \log_2 q \rceil$. As in the paper [MP12], for the case $q = 2^k$,

$$\mathbf{S}_k = \begin{pmatrix} 2 & & & & \\ -1 & 2 & & & \\ & -1 & \ddots & & \\ & & & 2 & \\ & & & -1 & 2 \end{pmatrix} \in \mathcal{R}^{k\times k}$$

and for $q < 2^k$,

$$\mathbf{S}_k = \begin{pmatrix} 2 & & & & q_0 \\ -1 & 2 & & & q_1 \\ & -1 & & & q_2 \\ & & \ddots & & \vdots \\ & & & 2 & q_{k-2} \\ & & & -1 & q_{k-1} \end{pmatrix} \in \mathcal{R}^{k\times k}$$

is a basis of the ring G-lattice $\Lambda_q^{\perp}(\mathbf{g}^{\top}) \subset \mathcal{R}^k$ and $(q_0, q_1, q_2, \ldots, q_{k-2}, q_{k-1}) \in \{0,1\}^k$ is the binary representation of $q = \sum_i 2^i \cdot q_i$. The module G-lattice is $\Lambda_q^{\perp}(\mathbf{G}) \subset \mathcal{R}^{dk}$, and it decomposes as

$$\Lambda_q^{\perp}(\mathbf{G}) \simeq \bigoplus_{i=1}^{d} \Lambda_q^{\perp}(\mathbf{g}^{\top}).$$

Note that,

$$\mathbf{S} = \begin{pmatrix} \mathbf{S}_k & & & \\ & \mathbf{S}_k & & \\ & & \ddots & \\ & & & \mathbf{S}_k \end{pmatrix} \in \mathcal{R}^{dk \times dk}$$

is a basis of $\Lambda_q^{\perp}(\mathbf{G})$. By the direct sum construction, sampling $\Lambda_q^{\perp}(\mathbf{G})$ can be done by performing d sampling in parallel over ring G-lattice $\Lambda_q^{\perp}(\mathbf{g}^{\top})$ and concatenating the results. We have already defined the G-lattice; now we can define G-trapdoor.

Definition 8. *Let m=$2d$+dk and $\boldsymbol{A} \in \mathcal{R}_q^{d \times m}$. A matrix $\boldsymbol{R} \in \mathcal{R}_q^{(m-dk) \times dk}$ is called a G-trapdoor of $\boldsymbol{A}$ if $\boldsymbol{A} \begin{bmatrix} \boldsymbol{R} \\ \boldsymbol{I}_{dk} \end{bmatrix} = \boldsymbol{G}$. The quality of the trapdoor is measured by its singular value $s_1(\boldsymbol{R})$.*

Input : A security-bit λ, a Gaussian parameter $\sigma \geq \eta_{\varepsilon}(\mathbb{Z}^{2nd})$ where $\epsilon = 2^{-\lambda}$, a tail-cut bound t.
Output: $\mathbf{A} \in \mathcal{R}_q^{d \times m}$, a G-trapdoor $\mathbf{R} \in \mathcal{R}_q^{2d \times dk}$ with $\|\mathbf{R}\| \leq t\sigma\sqrt{2nd}$.

$\hat{\mathbf{A}} \xleftarrow{\$} \mathcal{R}_q^{d \times d}$;
$\bar{\mathbf{A}} := [\mathbf{I}_d \mid \hat{\mathbf{A}}] \in \mathcal{R}_q^{d \times 2d}$;
$\mathbf{R} \leftarrow \mathrm{D}_{\mathcal{R}^{2d \times dk}, \sigma}$;
$\mathbf{A} \leftarrow [\bar{\mathbf{A}} \mid \mathbf{G} - \bar{\mathbf{A}}\mathbf{R}] \in \mathcal{R}_q^{d \times m}$;
return $(\mathbf{A}, \mathbf{R})$

Algorithm 1: TrapGen

Theorem 2 ([BEP+21])**.** *Let q and d be positive integers, $k = \lceil \log_2 q \rceil$, $\sigma > 0$, and $(\boldsymbol{A}, \boldsymbol{R}) \leftarrow$* TrapGen$(\lambda, \sigma)$. *Then $\boldsymbol{R}^{2d \times dk}$ is a G-Trapdoor of $\boldsymbol{A}$ and $\boldsymbol{A}^{d \times m}$ is computationally indistinguishable from uniform based on the hardness of decision* Module-LWE$_{n,d,q,\sigma}$.

3.1 Gaussian Preimage Sampling

In this section, we describe for a matrix $\mathbf{A} \in \mathcal{R}_q^{d \times m}$ and $\mathbf{u} \in \mathcal{R}^d$, how to sample the lattice coset $\Lambda_q^{\mathbf{u}}(\mathbf{A}) = \{\mathbf{x} \in \mathcal{R}^m : \mathbf{A}\mathbf{x} = \mathbf{u} \pmod q\}$ in two steps: sampling an appropriate coset of the G-lattice followed by a perturbation sampling.

3.1.1 Sampling a coset of the G-lattice

We describe a sampling technique over $\Lambda_q^{u}(\mathbf{g}^{\top}) \subset \mathbb{Z}^k$ for the case $q < 2^k$, following [GM17]. Rather than sampling directly from $\Lambda_q^{u}(\mathbf{g}^{\top})$, the trapdoor basis $\mathbf{S}_k$ for the lattice is decomposed as the product of an invertible linear transformation $\mathbf{T}$ and a simple, structured, sparse upper-triangular matrix $\mathbf{D}$.

A vector is then sampled from a spherical discrete Gaussian distribution with parameter $\alpha > 0$ over a coset of the lattice $\Lambda(\mathbf{D})$, where $\mathbf{G}$ denotes the gadget matrix associated with $\mathbf{g}^{\top}$. The sampled vector is subsequently mapped back to the target lattice by applying the linear transformation $\mathbf{T}$. Although the distribution over $\Lambda(\mathbf{D})$ is spherical, the application of $\mathbf{T}$ results in a distribution with covariance $\alpha^2\mathbf{T}\mathbf{T}^\top$.

This issue is addressed using the Gaussian convolution technique introduced in [Pei10]. In particular, a perturbation vector $\mathbf{p}$ is sampled from a discrete Gaussian distribution with covariance $s^2\mathbf{I} - \alpha^2\mathbf{T}\mathbf{T}^\top$ and added to the transformed sample. Consequently, the final output distribution has covariance

$$\alpha^2\mathbf{T}\mathbf{T}^\top + \left(s^2\mathbf{I} - \alpha^2\mathbf{T}\mathbf{T}^\top\right) = s^2\mathbf{I},$$

as required. The linear transformation $\mathbf{T}$ and the sparse matrix $\mathbf{D}$ has the form:

$$\mathbf{T} = \begin{pmatrix} 2 & & & & \\ -1 & 2 & & & \\ & -1 & \ddots & & \\ & & & 2 & \\ & & & -1 & 2 \end{pmatrix} \quad \text{and} \quad \mathbf{D} = \begin{pmatrix} 1 & & & & d_0 \\ & 1 & & & d_1 \\ & & \ddots & & \vdots \\ & & & 1 & d_{k-2} \\ & & & & d_{k-1} \end{pmatrix}$$

where $d_i = \frac{d_{i-1}+q_i}{2}, i \in \{0, 1, \ldots, k-1\}, d_{-1} = 0$, $(q_0, \ldots, q_{k-1}) \in \{0,1\}^k$. We give a simple high-level algorithmic form of $\Lambda_q^u(\mathbf{g}^\top)$ sampling in Algorithm 2. For actual algorithms, refer to the [GM17, Figure 2].

Input : A security bit λ, a spherical discrete Gaussian parameter $\alpha \geq \max\left(10\sqrt{\log(2k(1+2^\lambda))/\pi}, 3\eta_{2^{-\lambda}}(\mathbb{Z})\right)$, $\mathbf{u} = [u]_2^k \in \mathbb{Z}_q$

Output: $\mathbf{v} \leftarrow \mathrm{D}_{\Lambda_q^u(\mathbf{g}^\top),\alpha}$

$\Sigma_1 := \mathbf{T}\mathbf{T}^\top$;
$b \geq s_1(\mathbf{T}\mathbf{T}^\top)$;
$\Sigma_2 := b^2\mathbf{I} - \Sigma_1 = \mathbf{L} \cdot \mathbf{L}^\top$; `// By Cholesky Decomposition`
$\mathbf{p} \leftarrow \mathrm{D}_{\Lambda(\Sigma_2),\frac{\alpha}{3}\mathbf{L}}$; `// Perturbation sampling`
$\mathbf{c} = \mathbf{T}^{-1}(\mathbf{u} - \mathbf{p})$;
$\mathbf{z} \leftarrow \mathrm{D}_{\Lambda(\mathbf{D}),-\mathbf{c},\frac{\alpha}{3}}$;
return $\mathbf{v} = \mathbf{u} + \mathbf{T}\mathbf{z}$

Algorithm 2: G sampling

Performing module G-sampling enables us to calculate a vector $\mathbf{w} \in \Lambda_q^{\mathbf{u}}(\mathbf{A})$ along with a G-trapdoor $\mathbf{R}$ for a given syndrome $\mathbf{u} \in \mathcal{R}^d$. We sample $\mathbf{z} \leftarrow \mathrm{D}_{\Lambda_q^{\mathbf{u}}(\mathbf{G}),\alpha}$ and ouput $\mathbf{x} = \begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix}\mathbf{z}$ as

$$\mathbf{A}\mathbf{x} = \mathbf{A}\begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix}\mathbf{z} = \mathbf{G}\mathbf{z} = \mathbf{u}.$$

But, $\mathbf{x}$ follows a non-spherical discrete Gaussian with a covariance matrix $\alpha^2\begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix}\begin{bmatrix}\mathbf{R}^\top & \mathbf{I}\end{bmatrix}$ that leaks information about the G-trapdoor $\mathbf{R}$. This problem is taken care of using the Gaussian convolution technique.

3.1.2 Perturbation Sampling

To sample a vector $\mathbf{w}$ from a spherical Gaussian over $\Lambda_q^{\mathbf{u}}(\mathbf{A})$, a perturbation vector is sampled from $\mathrm{D}_{\mathcal{R}^m,\sqrt{\Sigma_\mathbf{p}}}$ where $\Sigma_\mathbf{p} = s^2\mathbf{I} - \alpha^2\begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix}\begin{bmatrix}\mathbf{R}^\top & \mathbf{I}\end{bmatrix} \in \mathcal{P}^{m\times m}$. Then $\mathbf{x} = \begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix}\mathbf{z}$ is sampled from the module G-lattice for an adjusted syndrome $\mathbf{v} = \mathbf{u} - \mathbf{A}\mathbf{p}$ and output $\mathbf{w} = \mathbf{x} + \mathbf{p}$ and $\mathbf{w}$ has covariance $\Sigma_p + \alpha^2 \begin{bmatrix} \mathbf{R} \\ \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{R}^\top & \mathbf{I} \end{bmatrix} = s^2\mathbf{I}$, which does not leak any information about the trapdoor $\mathbf{R}$. The core of perturbation sampling lies in the ring $\mathcal{P}$ rather than the usual ring $\mathcal{R}$. As in most discrete Gaussian sampling algorithms, computations are done with real numbers, even if the end result is composed of integers only. Since $\mathcal{R}$ can be embedded in $\mathcal{P}$, sometimes we can consider entries of covariance matrices in $\mathcal{P}$.

Theorem 3 ([BEP+21, Lemma 7])**.** *Let* $\boldsymbol{R} \in \mathcal{P}^{2d\times dk}$, $s > 0$, $\alpha > 0$ *and* $\Sigma_{\boldsymbol{p}} = s^2\boldsymbol{I} - \alpha^2 \begin{bmatrix} \boldsymbol{R} \\ \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{R}^\top & \boldsymbol{I} \end{bmatrix} \in \mathcal{P}^{m\times m}$. *For* $0 < \epsilon \leq \frac{1}{2}$, *and if*

$$s > \max\left(\sqrt{(\alpha^2+1)s_1^2(\boldsymbol{R}) + \eta_\epsilon^2(\mathbb{Z}^{mn})}, \sqrt{\alpha^2 + \eta_\epsilon^2(\mathbb{Z}^{mn})}\right),$$

then we have $\Sigma_p > \eta_\epsilon^2(\mathbb{Z}^{mn})$. *Moreover, because of* $\Sigma_p > \eta_\epsilon^2(\mathbb{Z}^{mn})$, *Algorithm 3 returns a vector* $\boldsymbol{p} \in \mathcal{R}^m$ *whose distribution is statistically close to* $\mathrm{D}_{\mathcal{R}^m, \sqrt{\Sigma_p}}$.

Input : G-Trapdoor $\mathbf{R} \in \mathcal{R}^{2d\times dk}$, $s > 0$, $\alpha > 0$
Output: $\mathbf{p} \leftarrow \mathrm{D}_{\mathcal{R}^m, \sqrt{\Sigma_p}}$

$\mathbf{p}_r \leftarrow \mathrm{D}_{\mathcal{R}^{dk}, s^2-\alpha^2}$;
$\mathbf{c} \leftarrow \frac{-\alpha^2}{s^2-\alpha^2}\mathbf{R}\mathbf{p}_\ell$;
$\Sigma \leftarrow s^2\mathbf{I} - (\alpha^{-2} - s^{-2})^{-1}\mathbf{R}\mathbf{R}^\top$;
for $i = 2d-1, \ldots, 0$ **do**
  $\Sigma = \left[\begin{array}{c|c} \Sigma_1 & \Sigma_2 \\ \hline \Sigma_2^\top & f \end{array}\right]$; // $\Sigma_1 \in \mathcal{P}^{i\times i}, \Sigma_2 \in \mathcal{P}^{i\times 1}, f \in \mathcal{P}$
  $\mathbf{c} = (\mathbf{c}', c_i)$; // $\mathsf{c}' \in \mathcal{P}^i, c \in \mathcal{P}$
  $p_i \leftarrow \mathrm{D}_{\mathcal{R}, \sqrt{f}, c_i}$;
  $\mathbf{c} \leftarrow \mathbf{c}' + f^{-1}\Sigma_2(p_i - c_i)$; // $\mathsf{c} \in \mathcal{P}^i$
  $\Sigma \leftarrow \Sigma_1 - f^{-1}\Sigma_2\Sigma_2^\top$; // $\Sigma \in \mathcal{P}^{i\times i}$
end
$\mathbf{p}_\ell \leftarrow (p_0, p_1, \ldots, p_{2d-1})$;
$\mathbf{p} \leftarrow (\mathbf{p}_\ell, \mathbf{p}_r)$;
return p

Algorithm 3: PerturbSample

Theorem 4 ([BEP+21, Theorem 3])**.** *Let* $\boldsymbol{R} \in \mathcal{P}^{2d\times dk}$ *be a G-trapdoor for a matrix* $\boldsymbol{A} \in \mathcal{R}_q^{d\times m}$, $\boldsymbol{u} \in \mathcal{R}_q^d$ *be a syndrome,* $\alpha > 10\sqrt{\log(2k(1+2^\lambda))/\pi}$ *and* $s > \sqrt{(\alpha^2+1)s_1^2(\boldsymbol{R}) + \eta_\epsilon^2(\mathbb{Z}^{mn})}$ *be Gaussian parameters. Then Algorithm 4 outputs a vector whose distribution is statistically close to* $\mathrm{D}_{\Lambda_q^{\boldsymbol{u}}(\boldsymbol{A}), s}$.

4 Two-Round ID-based blind signature from G-trapdoor

We modify the trapdoor delegation algorithm DelTrap of [MP12] over module lattices. The modifications and correctness are quite straightforward.

Input : $\mathbf{A} \in \mathcal{R}_q^{d\times m}$, G-Trapdoor $\mathbf{R} \in \mathcal{R}^{2d\times dk}$, $\mathbf{u} \in \mathcal{R}_q^d$, $s > 0$, $\alpha > 0$
Output: $\mathbf{x} \leftarrow \mathrm{D}_{\Lambda_q^{\mathbf{u}}(\mathbf{A}),s}$

$\mathbf{p} \leftarrow \mathrm{D}_{\mathcal{R}^m,\sqrt{\Sigma_p}}$; `// using PerturbSample`
$\mathbf{v} \leftarrow \mathbf{u} - \mathbf{A}\mathbf{p}$;
$\mathbf{z} \leftarrow \mathrm{D}_{\Lambda_q^{\mathbf{v}}(\mathbf{G}),\alpha}$;
$\mathbf{x} \leftarrow \mathbf{p} + \begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix}\mathbf{z}$;
return x

Algorithm 4: SamplePre

Input : $\mathbf{A}' = [\mathbf{A} \mid \mathbf{A}_1] \in \mathcal{R}_q^{d\times m} \times \mathcal{R}_q^{d\times dk}$, A Gaussian parameters s.
Output: A G-Trapdoor $\mathbf{R}' \in \mathcal{R}^{m\times dk}$ for $\mathbf{A}'$.

$\mathbf{G} - \mathbf{A}_1 = [\mathbf{g}'_1, \ldots, \mathbf{g}'_{dk}]$;
for $i = 1, \ldots, dk$ **do**
  $\mathbf{r}'_i \leftarrow \mathrm{D}_{\Lambda_q^{\mathbf{g}'_i}(\mathbf{A}),s}$; `// Using algorithm SamplePre`
end
return $\mathbf{R}' = [\mathbf{r}'_1, \ldots, \mathbf{r}'_{dk}]$

Algorithm 5: DelTrap

If $\mathbf{R}'$ is unstructured over $\mathbb{R}$, we could apply Lemma 2. 9 of [MP12], which states that $s_1(\mathbf{R}') < C \cdot s \cdot (\sqrt{nm} + \sqrt{ndk} + c)$ for a universal constant $C > 0$, except with a probability at most $2\exp(-\pi c^2)$. The $\mathbf{R}'$ in our case is structured, but as claimed to be empirically confirmed by the paper [BEP+21], we assume that the lemma can be applied for the structured case also. We choose $c = 4.7$ so that $2\exp(-\pi c^2) \approx 2^{-100}$. For discrete Gaussian, the universal constant C is very close to $1/\sqrt{2\pi}$. Heuristically, we have chosen $C \approx 1/\sqrt{2\pi}$.

4.1 The Unblinded GPV Signature scheme

The GPV signature scheme [BEP+21, Appendix B] is EU-CMA secure in the random oracle model under the hardness of Module-MLWE and Module-SIS. However, the scheme is not competitive, in terms of efficiency and signature size, with the state-of-the-art constructions such as [PFH+17, DKL+18].

We modify the signature scheme to meet our requirements through a two-fold extension. First, we introduce a public matrix $\mathbf{A} \in \mathcal{R}_q^{d\times m}$ together with a corresponding G-trapdoor $\mathbf{R}^{2d\times dk}$ and an identity string ID. Public parameters $\mathrm{pp} = (\lambda, \mathbf{G}, \mathbf{B}, \mathbf{A}_1, \mathbf{A}_2, \mathcal{H}, \mathrm{H})$ where λ is a security parameter and $\mathbf{B} \in \mathcal{R}_q^{d\times m}$, $\mathbf{A}_1 \in \mathcal{R}_q^{d\times m}$ and $\mathbf{A}_2 \in \mathcal{R}_q^{d\times 1}$ are non-trapdoored uniform matrices, $\mathrm{H} : \{0,1\}^\star \to \mathcal{R}_q^d$, $\mathcal{H} : \{0,1\}^\star \to \mathcal{R}_q^{d\times dk}$ a re cryptographic hash functions. The signer hashes ID to obtain a matrix $\mathcal{H}(\mathrm{ID}) \in \mathcal{R}_q^{d\times dk}$ and constructs

$$\mathbf{A}_{\mathrm{ID}} = [\mathbf{A} \mid \mathcal{H}(\mathrm{ID})] \in \mathcal{R}_q^{d\times(m+dk)}.$$

The signing key is a G-trapdoor $\mathbf{R}_{\mathrm{ID}} \in \mathcal{R}_q^{m\times dk}$ for $\mathbf{A}_{\mathrm{ID}}$, which makes it possible to sample a short preimage $\mathbf{y} \in \mathcal{R}_q^{m+dk}$ by using the SamplePre algorithm, satisfying

$$\mathbf{A}_{\mathrm{ID}}\mathbf{y} = \mathrm{H}(\mu),$$

where the hash function H is modelled as a random oracle mapping into $\mathcal{R}_q^d$. SamplePre uses PerturbSample as a subroutine. In the basic signature algorithm, PerturbSample initially uses the matrix

$$\Sigma = s'^2\mathbf{I} - (\alpha^{-2} - s'^{-2})^{-1}\mathbf{R}_{\mathrm{ID}}\mathbf{R}_{\mathrm{ID}}^\top$$

of dimension $m \times m$ and so the component $\mathbf{p}_\ell$ of the perturbation vector $\mathbf{p}$ is $(p_0, p_1, \ldots, p_{m-1})$.

Second, we extend the construction to an unblinded signature scheme by introducing a user-chosen short vector $\mathbf{x} \in \mathcal{R}_q^m$ and a non-trapdoored public matrix $\mathbf{B} \in \mathcal{R}^{d\times m}$. A signature on a message μ under identity ID consists of a Gaussian preimage $\mathbf{y}$ and an Ajtai commitment $\mathrm{com}_\mathbf{x}$ to a short vector $\mathbf{x}$ such that

$$\mathbf{A}_{\mathrm{ID}}\mathbf{y} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_\mathbf{x}, \mu, \mathrm{ID}),$$

where $\mathrm{com}_\mathbf{x}$ denotes an Ajtai commitment to $\mathbf{x}$. Including ID in the hash input ensures that the hashing is performed for a particular ID, while in the case of a blind signature, it ensures that the message μ is being blinded for a blind signer consisting of an ID.

Replacing the hash of $\mathbf{x}$ (as in [BLNS23]) with an Ajtai commitment removes the reliance on the collision resistance and random-oracle programmability as we discussed in subsection 1.1. As a result, the security reduction relies on the computational binding property of the commitment, yielding a cleaner and more robust extraction argument based on Module-SIS. The vector $\mathbf{t} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_\mathbf{x}, \mu, \mathrm{ID})$ is pseudorandom by "one time pad" style mechanism. This randomness is essential in the security game, where the adversary simulates the signing oracle by responding to signing queries with uniformly random vectors $\mathbf{t}$. Finally, note that no NIZK opening proof for the Ajtai commitment is required during verification in this unblinded version, since we take the witnesses $(\mathbf{x}, \mathbf{r})$ as inputs here; and any successful forgery yields a solution to Module-SIS for the matrix $(\mathbf{A}_{\mathrm{ID}} \mid -\mathbf{B})$ and syndrome $\mathrm{H}(\mathrm{com}_\mathbf{x}, \mu, \mathrm{ID})$.

Gen(λ)

1. $([\bar{\mathbf{A}} \mid \mathbf{G} - \bar{\mathbf{A}}\mathbf{R}], \mathbf{R}) \leftarrow \mathrm{TrapGen}(\lambda, \sigma)$;
2. Set $mpk_{\mathrm{IBBS}} = \mathbf{A} = [\bar{\mathbf{A}} \mid \mathbf{G} - \bar{\mathbf{A}}\mathbf{R}]$, $msk_{\mathrm{IBBS}} = \mathbf{R}$;
3. Generate $\mathbf{A}_1 \xleftarrow{\$} \mathcal{R}_q^{d\times m}$, $\mathbf{A}_2 \xleftarrow{\$} \mathcal{R}_q^{d\times 1}$, $\mathbf{B} \xleftarrow{\$} \mathcal{R}_q^{d\times m}$;
4. Construct the cryptographic hash functions $\mathrm{H} : \{0,1\}^\star \to \mathcal{R}_q^d$, $\mathcal{H} : \{0,1\}^\star \to \mathcal{R}_q^{d\times dk}$;
5. A hash function $\mathcal{H}_{\mathrm{PKE}} : \{0,1\}^\star \to \mathcal{K}_{\mathrm{PKE}}$;
6. $\mathrm{pp} = (\lambda, \mathbf{G}, \mathbf{B}, \mathbf{A}_1, \mathbf{A}_2, \mathcal{H}, \mathrm{H}, \mathcal{H}_{\mathrm{PKE}})$;
7. Return $\mathrm{pp}, mpk_{\mathrm{IBBS}}, msk_{\mathrm{IBBS}}$.

Extract$(\mathrm{ID} \in \mathcal{ID}, \mathrm{pp}, mpk_{\mathrm{IBBS}}, msk_{\mathrm{IBBS}}, s > 0, \alpha > 0)$

1. Compute $\mathrm{H}(\mathrm{ID}) \in \mathcal{R}_q^{d\times dk}$;
2. Set $\mathbf{A}_{\mathrm{ID}} := [\mathbf{A} \mid \mathrm{H}(\mathrm{ID})] \in \mathcal{R}^{d\times(m+dk)}$;
3. $\mathbf{R}_{\mathrm{ID}} \leftarrow \mathrm{DelTrap}(\mathbf{A}_{\mathrm{ID}}, \mathbf{G}, s)$;
4. Output $(\mathbf{A}_{\mathrm{ID}}, \mathbf{R}_{\mathrm{ID}})$

Sign$(\mathrm{ID}, \mathrm{Sig.vk} = \mathbf{A}_{\mathrm{ID}}, \mathrm{Sig.sk} = \mathbf{R}_{\mathrm{ID}}, \mu \in \{0,1\}^\star, \mathrm{pp}, \mathbf{x} \leftarrow \mathrm{D}_{\mathcal{R}^m,\sigma_\mathbf{x}}, \mathbf{r} \leftarrow \mathrm{D}_{\mathcal{R},\sigma_\mathbf{r}}, s' > 0, \alpha > 0)$

1. $\|\mathbf{x}\| > \beta_\mathbf{x}$ or $\|\mathbf{r}\| > \beta_\mathbf{r}$, return $\perp$;
2. Generate Ajtai commitment $\mathrm{com}_\mathbf{x} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$;

3. Compute $\mathbf{t} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$;
4. $\mathbf{y} \leftarrow \mathrm{SamplePre}(\mathbf{R}_{\mathrm{ID}}, \mathbf{A}_{\mathrm{ID}}, \mathbf{t}, s', \alpha)$;
5. Output $\mathrm{sig} = (\mathrm{com}_{\mathbf{x}}, \mathbf{y})$

Verify$(\mu, (\mathbf{x}, \mathbf{r}), \mathrm{sig}, \mathrm{ID}, \mathrm{pp})$

1. Parse $\mathrm{sig} = (\mathrm{com}_{\mathbf{x}}, \mathbf{y})$;
2. If $\|\mathbf{y}\| > \beta_{\mathbf{y}}$ or $\|\mathbf{x}\| > \beta_{\mathbf{x}}$ or $\|\mathbf{r}\| > \beta_{\mathbf{r}}$, output $\perp$;
3. If $\mathrm{com}_{\mathbf{x}} \neq \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$, output $\perp$;
4. If $\mathbf{A}_{\mathrm{ID}}\mathbf{y} \neq \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$; output $\perp$;
5. Output Verified

Figure 1: Unblinded GPV Signature Scheme

ID-based Blind Signature Scheme Here, we extend the modified unblinded GPV signature described above into our final ID-based blind signature scheme. The signature scheme initiates with the trusted authority (TA) setting up the public parameters (pp) and generating the master public and secret keys as in Fig. 2. To begin with, it generates a collection of polynomially many IDs and assigns them to a group of blind signers of its choice. The master public key $\mathbf{A}^{d\times m}$ is generated using the TrapGen along with its corresponding G-trapdoor $\mathbf{R}^{2d\times dk}$, which is kept as the master secret key. The TA also samples non-trapdoored uniform matrices $\mathbf{B}$, $\mathbf{A}_1$, $\mathbf{A}_2$ which are published as part of the public parameters along with the public matrix $\mathbf{G}$, and the security parameter λ.

Now, when the user wants a blind signature on a message μ, it first blinds the message for a particular ID as in Fig. 3. The user samples a short vector $\mathbf{x}$ and computes an Ajtai commitment $\mathrm{com}_{\mathbf{x}}$. Then it constructs the blinded message $\mathbf{t} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$. The user also computes the encryption public key by $\mathrm{PKE.pk} = \mathcal{H}_{\mathrm{PKE}}(\mathrm{ID})$ and encrypts the witnesses $(\mathbf{x}, \mathbf{r}, \mu)$ as the ciphertext ct. Finally, along with $\mathbf{t}$ and ct, an NIZK proof π_1 of correct computation of the ciphertext is send to the blind signer having that specific ID.

The signer samples a short preimage $\mathbf{y}$ and returns it to the user, who can then prove knowledge of a valid signature along with an opening proof of the $\mathrm{com}_{\mathbf{x}}$. Revealing the commitment $\mathrm{com}_{\mathbf{x}}$ with the NIZK proof π_2 fixes the value of $\mathbf{h} = \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$, while the term $\mathbf{x}$ and $\mathbf{r}$ remain hidden due to the computational hiding property of the Ajtai commitment. Moreover, the contribution of $\mathbf{B}\mathbf{x}$ is that it statistically masks $\mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$ ensuring that $\mathbf{t}$ remains unpredictable to the signer.

Theorem 5. *The basic signature scheme (Figure 1) is one-more unforgeable under the* MLWE *and* MSIS *assumptions when the hash function* H *is modelled as a random oracle, i.e., if a forger $\mathcal{A}$ in the signature forgery game, queries at most Q_{H} times to the random oracle and has advantage $\mathrm{Adv}_{SIG}(\mathcal{A})$, then there exist an adversary $\mathcal{B}$ that distinguishes the* Module-LWE *master public-key from uniform with advantage $\mathrm{Adv}_{\mathrm{MLWE}}(\mathcal{B})$, an adversary $\mathcal{C}$ that solves* Module-SIS *with norm bound $\beta = 2\sqrt{\beta_x^2 + \beta_y^2}$ with advantage $\mathrm{Adv}_{\mathrm{MSIS}_\beta}(\mathcal{C})$ and another adversary $\mathcal{D}$ with advantage $\mathrm{Adv}_{\mathrm{BIND}}(\mathcal{D})$ such that*

$$\mathrm{Adv}_{\mathrm{SIG}}(\mathcal{A}) \leq \mathrm{Adv}_{\mathrm{MLWE}}(\mathcal{B}) + \mathrm{Adv}_{\mathrm{MSIS}_\beta}(\mathcal{C}) + \mathrm{Adv}_{\mathrm{BIND}}(\mathcal{D}) + Q_{\mathrm{H}}/q^{nd}$$

Proof. Let $\mathcal{A}$ be a polynomial-time adversary. We prove that $\mathcal{A}$ has a negligible probability of winning the one-more unforgeability game using a sequence of games.

Trusted Authority | **Signer**

1. Creates $\text{pp} = (\lambda, \mathbf{G}, \mathbf{B}, \mathbf{A}_1, \mathbf{A}_2, \mathcal{H}, \mathrm{H}, \mathcal{H}_{\text{PKE}})$
2. Generates $\mathcal{ID} = \{\text{ID}_i : i \in [1, N]\}$.
3. Assigns the IDs to the blind signers.
4. $\bar{\mathbf{A}} \xleftarrow{\$} \mathcal{R}_q^{d \times 2d}$
5. $(\mathbf{A}, \mathbf{R}) \leftarrow \text{TrapGen}(1^\lambda)$ where $\mathbf{A} = [\bar{\mathbf{A}} \mid \mathbf{G} - \bar{\mathbf{A}}\mathbf{R}]$
6. $mpk = \mathbf{A} \in \mathcal{R}_q^{d \times m}$
7. $msk = \mathbf{R} \in \mathcal{R}^{2d \times dk}$

Signer: Assigned some id, ID_k for some $k \in [1, N]$

$\xleftarrow{\text{ID}_k}$

Trusted Authority:

$\mathbf{A}_{\text{ID}_k} = [\mathbf{A} \mid H(\text{ID}_k)] \in \mathcal{R}_q^{d \times (m+dk)}$
$\mathbf{R}_{\text{ID}_k} \leftarrow \text{DelTrap}(\mathbf{A}_{\text{ID}_k}, s)$

$\xrightarrow{\mathbf{A}_{\text{ID}_k}, \mathbf{R}_{\text{ID}_k}}$

Signer:

$\text{IDBSig.vk} = \mathbf{A}_{\text{ID}_k}$
$\text{IDBSig.sk} = \mathbf{R}_{\text{ID}_k}$

Figure 2: Key Generation

- $\mathbf{Game_0}$: This is the real *one-more unforgeability* game for an ID.
- $\mathbf{Game_1}$: This is similar to $\mathbf{Game_0}$ except in this game the hash function is programmed. We describe a simulator for an ID that answers hash and signing queries without knowledge of the trapdoor. The simulator maintains the following relations:
  - $R = \{(\mathbf{x}, \mathbf{r}, \mathbf{v})\}$
  - $S = \{(\mathbf{v}, \mu, \mathbf{h})\}$
  - $S' = \{(\mathbf{x}, \mathbf{y}, \mathbf{h})\}$

  where $\mathbf{v} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$ is an Ajtai commitment to $\mathbf{x}$ with randomness $\mathbf{r}$.

  Hash queries. For a hash query on $(\mathbf{v}, \mu)$, if $(\mathbf{v}, \mu, \mathbf{h}) \in S$, simulator returns $\mathbf{h} \leftarrow \mathrm{H}(\mathbf{v}, \mu)$. Else, if there exits $(\mathbf{x}, \mathbf{r}, \mathbf{v}) \in R$ with $\|\mathbf{x}\| \leq \beta_{\mathbf{x}}$, choose $\mathbf{x}$ or, $\mathbf{x} \leftarrow \mathrm{D}_{\mathcal{R}^m, \sigma_{\mathbf{x}}}$, $\mathbf{y} \leftarrow \mathrm{D}_{\mathcal{R}^{(m+dk)}, s'}$ and programmes

  $$\mathbf{h} := \mathbf{A}_{\text{ID}}\mathbf{y} - \mathbf{B}\mathbf{x}.$$

  The simulator stores $S = S \cup \{(\mathbf{v}, \mu, \mathbf{h})\}$ and $S' = S' \cup \{(\mathbf{x}, \mathbf{y}, \mathbf{h})\}$. Since $\sigma \geq \eta_\epsilon(\Lambda_q^\perp(\mathbf{A}_{\text{ID}}))$, the distribution of $\mathbf{h}$ is statistically close to uniform by Lemma 3.

  Signing queries. For a signing query on $(\mu, \mathbf{x}, \mathbf{r})$, the simulator retrieves $\mathbf{v}$ from R or computes

  $$\mathbf{v} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$$

  where $\mathbf{x} \leftarrow \mathrm{D}_{\mathcal{R}^m, \sigma_{\mathbf{x}}}$, $\mathbf{r} \leftarrow \mathrm{D}_{\mathcal{R}, \sigma_{\mathbf{r}}}$ and stores $R = R \cup \{(\mathbf{x}, \mathbf{r}, \mathbf{v})\}$. Next, the simulator queries the hash oracle on $(\mathbf{v}, \mu)$. The hash oracle gives $\mathbf{h} \leftarrow \mathrm{H}(\mathbf{v}, \mu)$. Then the relation S' contains a preimage $(\mathbf{x}', \mathbf{y})$ such that $\mathbf{A}_{\text{ID}}\mathbf{y} - \mathbf{B}\mathbf{x}' = \mathbf{h}$ and for $\mathbf{x}'$ there exists $(\mathbf{x}', \mathbf{r}', \mathbf{v}) \in R$.

Public Parameter: $\mathrm{pp} = (\lambda, \mathbf{G}, \mathbf{B}, \mathbf{A}_1, \mathbf{A}_2, \mathcal{H}, \mathrm{H}, \mathcal{H}_{\mathrm{PKE}})$

Signer($\mathcal{S}$)		User($\mathcal{U}$)
$\mathrm{IDBSig.sk} = \mathbf{R}_{\mathrm{ID}} \in \mathcal{R}^{m \times dk}$		$\mathrm{IDBSig.vk} = \mathrm{ID}$
		1. Sample $\mathbf{x} \leftarrow \mathrm{D}_{\mathcal{R}^m, \sigma_{\mathbf{x}}}$ and $\mathbf{r} \leftarrow \mathrm{D}_{\mathcal{R}, \sigma_{\mathbf{r}}}$; 2. $\|\mathbf{x}\| > \beta_{\mathbf{x}}$ or $\|\mathbf{r}\| > \beta_{\mathbf{r}}$, return $\perp$; 3. $\mathrm{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r} \in \mathcal{R}_q^d$; 4. $\mathbf{t} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$; 5. $\mathrm{PKE.pk} = \mathcal{H}_{\mathrm{PKE}}(\mathrm{ID})$; 6. $\mathrm{ct} = \mathrm{PKE.enc}(\mathrm{PKE.pk}, \mathbf{x} \parallel \mathbf{r} \parallel \mu; \mathtt{rnd})$, $\mathtt{rnd} \xleftarrow{\$} \{0,1\}^{\lambda}$; 7. An NIZK proof π_1 proving correct computation of ct.
$\mathbf{y} \leftarrow \mathbf{A}_{\mathrm{ID}_{s}'}^{-1}(\mathbf{t})$ Resample if $\|\mathbf{y}\| > \beta_y$.	$\xleftarrow{\mathbf{t}, \pi_1, \mathrm{ct}}$ $\xrightarrow{\mathbf{y}}$	
		Construct NIZK proof π_2 for witnesses $\mathbf{x}, \mathbf{y}, \mathbf{r}$ s.t. 1. $\mathrm{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$; 2. $\mathbf{A}_{\mathrm{ID}} \cdot \mathbf{y} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$; 3. $\|\mathbf{x}\| \leq \beta_{\mathbf{x}}, \|\mathbf{y}\| \leq \beta_{\mathbf{y}}, \|\mathbf{r}\| \leq \beta_{\mathbf{r}}$; Sig. for μ is $(\mathrm{com}_{\mathbf{x}}, \pi_2)$.

Verification: Return 1 if π_2 is valid.

Figure 3: ID-Based Blind Signature Scheme

If $\mathbf{x} \neq \mathbf{x}'$ or $\mathbf{r} \neq \mathbf{r}'$, then the verification fails and the forgery game is aborted.

Now we examine the probability of the aborting condition. If the hash oracle outputs the same value $\mathbf{h}$ for two different inputs during signing queries

$$\mathrm{H}(\mathbf{v}_1, \mu_1) = \mathrm{H}(\mathbf{v}_2, \mu_2) = \mathbf{h}$$

with $(\mathbf{v}_1, \mu_1) \neq (\mathbf{v}_2, \mu_2)$, it means the relation S' contains distinct pairs $(\mathbf{x}_1, \mathbf{y}_1, \mathbf{h})$ and $(\mathbf{x}_2, \mathbf{y}_2, \mathbf{h})$. Also, R contains two distinct pairs $(\mathbf{x}_1, \mathbf{r}_1, \mathbf{v}_1)$ and $(\mathbf{x}_2, \mathbf{r}_2, \mathbf{v}_2)$. If the signing query corresponding to $\mathbf{x} = \mathbf{x}_1$ and $\mathbf{r} = \mathbf{r}_1$, but the simulator picks $(\mathbf{x}_2, \mathbf{y}_2)$ then $\mathbf{x}' = \mathbf{x}_2$ and $\mathbf{r}' = \mathbf{r}_2$, so the simulator aborts. Since $\mathbf{h}$ is uniformly distributed over $\mathcal{R}_q^d$, the probability of the collision is at most $Q_{\mathrm{H}}^2/q^{nd+1}$, which is negligible. On the other hand, suppose the adversary queries the hash oracle before the signing query. It can happen in two ways. First, during a hash query $\mathrm{H}(\mathbf{v}, \mu)$ with $\mathbf{v} = \mathbf{A}_1\mathbf{x}' + \mathbf{A}_2\mathbf{r}'$, the simulator does not know the opening $(\mathbf{x}', \mathbf{r}')$ of $\mathbf{v}$. It chooses an $\mathbf{x}'$ independently when programming $\mathbf{h}$. Thus it stores $(\mathbf{x}', \mathbf{y}, \mathbf{h}) \in S'$. Later during the signing query on $(\mu, \mathbf{x}, \mathbf{r})$, the simulator computes $\mathbf{v} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$. Next, on the input $(\mathbf{v}, \mu)$ to the hash oracle, the simulator retrieves the stored tuple $(\mathbf{x}', \mathbf{y}, \mathbf{h})$, which leads to an abort. The equality, for some $\mathbf{r}'$,

$$\mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r} = \mathbf{A}_1\mathbf{x}' + \mathbf{A}_2\mathbf{r}' \implies [\mathbf{A}_1|\mathbf{A}_2]\begin{pmatrix}\mathbf{x}-\mathbf{x}' \\ \mathbf{r}-\mathbf{r}'\end{pmatrix} = \mathbf{0}$$

yields a non-zero MSIS solution with the bound $2\sqrt{\beta_x^2 + \beta_r^2}$ with the advantage $\mathrm{Adv}_{\mathrm{BIND}}$.

Second, the adversary queries the hash oracle $\mathrm{H}(\mathbf{v}, \mu)$ by sampling $\mathbf{v}$ from the commitment space $\mathcal{R}_q^d$. The simulator stores $(\mathbf{x}, \mathbf{y}, \mathbf{h})$ and returns $\mathbf{h}$. Later on, signing query on μ, the simulator computes $\mathbf{v} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$ for fresh sampled $(\mathbf{x}, \mathbf{r})$. Then the simulator retrieves $(\mathbf{x}', \mathbf{y}, \mathbf{h}) \in S'$. If $\mathbf{x} \neq \mathbf{x}'$, then the simulator aborts.

Thus, the probability that the adversary guesses a future commitment is bounded by Q_{H}/q^{nd} for Q_{H} many hash queries. So the Pr[Abort]$\leq Q_{\mathrm{H}}/q^{nd} + \mathrm{Adv}_{\mathrm{BIND}}$ and hence the $\mathbf{Game_1}$ is at most $Q_{\mathrm{H}}/q^{nd} + \mathrm{Adv}_{\mathrm{BIND}}$ away from $\mathbf{Game_0}$.

- $\mathbf{Game_2}$: Same as $\mathbf{Game_1}$, except the master public key is replaced by a uniformly random one by the challenger. Then there exists an adversary $\mathcal{B}$ which can distinguish $\mathbf{Game_1}$ from $\mathbf{Game_2}$ with the advantage $\mathrm{Adv}_{\mathrm{MLWE}}(\mathcal{B})$.
- $\mathbf{Game_3}$: Same as $\mathbf{Game_2}$, except that upon a successful forgery by the adversary $\mathcal{A}$, there exists an adversary $\mathcal{C}$ with an advantage $\mathrm{Adv}_{\mathrm{MSIS}_\beta}(\mathcal{C})$ extracts a MSIS solution. Suppose the adversary outputs $Q_s + 1$ many valid message-signature pairs i.e. $\{(\mu_i, (\mathbf{x}_i, \mathbf{r}_i), (\mathrm{com}_{\mathbf{x}_i}, \mathbf{y}_i)\}_{i=1}^{Q_s+1}$. There exists at least one signature $(\mathrm{com}_{\mathbf{x}}, \mathbf{y})$ for a message μ that was not previously used for signing query such that

$$\mathbf{A}_{\mathrm{ID}}\mathbf{y} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID}).$$

By construction of the hash oracle, there exists a stored tuple $(\mathbf{x}', \mathbf{y}', \mathbf{h})$ in the relation R with

$$\mathbf{h} = \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID}) \quad \text{and} \quad \mathbf{A}_{\mathrm{ID}}\mathbf{y}' = \mathbf{B}\mathbf{x}' + \mathbf{h}.$$

From the lemma 2, and the fact $\sigma \geq \eta_\epsilon\left(\Lambda_q^{\perp}(\mathbf{A}_{\mathrm{ID}})\right)$, the conditional distribution of $\mathbf{y}'$ has very large entropy, so from the subsection 2.7, $\mathbf{y} = \mathbf{y}'$ with low probability. Therefore,

$$\mathbf{A}_{\mathrm{ID}}(\mathbf{y} - \mathbf{y}') - \mathbf{B}(\mathbf{x} - \mathbf{x}') = \mathbf{0}$$

yields a non-zero Module-SIS solution of norm at most

$$\beta = 2\sqrt{\beta_x^2 + \beta_y^2}.$$

$\square$

Theorem 6. *The ID-based blind signature scheme in Figure 3 is selective-ID one-more unforgeable, assuming the one-more unforgeability of the underlying signature scheme and the soundness of the NIZK proofs. More precisely, if there exists an adversary $\mathcal{A}$ making at most Q_s signing queries in the selective-ID one-more-unforgeability game, then there exist adversaries $\mathcal{B}$ and $\mathcal{C}$ such that*

$$\mathrm{Adv}_{\mathrm{IDBSIG}}(\mathcal{A}) \leq (Q_s + 1)\,\mathrm{Adv}_{\mathrm{SOUND}}(\mathcal{B}) + \mathrm{Adv}_{\mathrm{UBSIG}}(\mathcal{C}).$$

Proof. In the selective-ID game, the adversary $\mathcal{A}$ first commits to an identity $\mathrm{ID}^\star$. The challenger then samples the master public key mpk, master secret key msk and pp. Moreover, TA runs $(\mathrm{PKE.pk}, \mathrm{PKE.sk}) \leftarrow \mathrm{PKE.KeyGen}(\lambda)$, programs $\mathcal{H}_{\mathrm{PKE}}(\mathrm{ID}^\star) := \mathrm{PKE.pk}$ and keeps PKE.sk.

Key Extraction. The adversary may issue key extraction queries for any identity $\mathrm{ID} \neq \mathrm{ID}^\star$. The challenger responds by returning the corresponding signing key $\mathcal{S}_{\mathrm{ID}}$.

Signing Queries. To obtain a blind signature under identity $\mathrm{ID}^\star$, the adversary chooses a message μ, samples vectors $\|\mathbf{x}\| \leq \beta_{\mathbf{x}}$ and $\|\mathbf{r}\| \leq \beta_{\mathbf{r}}$, computes $\mathrm{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$, performs the blinding procedure, and sends to the challenger

$$\begin{aligned}
&\mathbf{t} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID}^\star);\\
&\mathrm{ct} = \mathrm{PKE.enc}(\mathrm{PKE.pk}, \mathbf{x} \parallel \mathbf{r} \parallel \mu; \mathtt{rnd}), \mathtt{rnd} \xleftarrow{\$} \{0,1\}^\lambda;\\
&\pi_1 : \left(\exists\, \mathbf{x}, \mathbf{r}, \mu, \mathtt{rnd} \;\middle|\; \begin{array}{l} \mathrm{ct} = \mathrm{PKE.enc}(\mathrm{PKE.pk}, \mathbf{x} \parallel \mathbf{r} \parallel \mu; \mathtt{rnd}) \wedge \\ \|\mathbf{x}\| \leq \beta_{\mathbf{x}} \wedge \|\mathbf{r}\| \leq \beta_{\mathbf{r}} \wedge \mathtt{rnd} \xleftarrow{\$} \{0,1\}^\lambda \end{array} \right)
\end{aligned}$$

The challenger verifies the proof π_1 then decrypts the ciphertext ct and queries the signing oracle with (μ, $\mathbf{x}$, $\mathbf{r}$). The hash oracle looks up the relation R with the input ($\mathrm{com}_{\mathbf{x}}$, μ) in the unforgeability game of the unblinded signature scheme, retrieves the signature $(\mathrm{com}_{\mathbf{x}}, \mathbf{y})$, and returns $\mathbf{y}$ to the adversary.

Forgery. Suppose, $\mathcal{A}$ outputs $(Q_s + 1)$ valid message-signature pairs $(\mu_i, (\mathrm{com}_{\mathbf{x}_i}, \pi_i))_{i=1}^{Q_s+1}$ under identity $\mathrm{ID}^\star$. By the soundness property of the NIZK, an adversary can not come up with a valid proof of a false statement, except with the advantage $\mathrm{Adv}_{\mathrm{SOUND}}(\mathcal{B})$. So assume that all the statements are true. Since only Q_s signing queries were answered, there exists a signed message $(\mu, (\mathrm{com}_{\mathbf{x}}, \pi_2))$ where the message μ is never used. One can then use an extractor of the NIZK to extract the witnesses ($\mathbf{x}$, $\mathbf{r}$, $\mathbf{y}$) such that

$$\mathbf{A}_{\mathrm{ID}^\star}\mathbf{y} = \mathbf{B}\mathbf{x} + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID}^\star)$$

and

$$\mathrm{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$$

with $\|\mathbf{y}\| \leq \beta_y$, $\|\mathbf{x}\| \leq \beta_x$, $\|\mathbf{r}\| \leq \beta_r$. Hence, $(\mathrm{com}_{\mathbf{x}}, \mathbf{y})$ constitutes a valid signature on the new message μ, yielding a one-more forgery for the underlying unblinded signature scheme. This completes the reduction. $\square$

Theorem 7. *Assume that NIZK is zero-knowledge. Then the ID-based blind signature in figure 3 satisfies honest signer blindness, i.e., for every adversary $\mathcal{A}$ in the honest signer blindness game there exists an adversary $\mathcal{B}$ that can distinguish simulated NIZK proofs from the real one with advantage* $\mathrm{Adv}_{\mathrm{ZK}}(\mathcal{B})$*, an adversary $\mathcal{C}$ against the CPA security of the encryption scheme with advantage* $\mathrm{Adv}_{\mathrm{CPA}}(\mathcal{C})$*, an adversary $\mathcal{D}$ that can distinguish Ajtai Commitment with the advantage* $\mathrm{Adv}_{\mathrm{HIDE}}(\mathcal{D})$ *such that*

$$\mathrm{Adv}_{\mathrm{HSB}}(\mathcal{A}) \leq \mathrm{Adv}_{\mathrm{ZK}}(\mathcal{B}) + \mathrm{Adv}_{\mathrm{CPA}}(\mathcal{C}) + \mathrm{Adv}_{\mathrm{HIDE}}(\mathcal{D})$$

Proof. We prove the theorem via a sequence of games.

- $\mathbf{Game_0}$: This is the real anonymity game. The challenger generates the keys and runs the blind signing protocol honestly on the input message μ_b, where $b \leftarrow \{0,1\}$ is chosen uniformly at random. The adversary is given the resulting transcript and outputs a bit b'.

- $\mathbf{Game_1}$: This game is similar to $\mathbf{Game_0}$, except that the NIZK proof in the protocol is replaced by simulated proofs. Then there exists an adversary $\mathcal{B}$ who can distinguish $\mathbf{Game_1}$ from $\mathbf{Game_0}$ with the advantage $\mathrm{Adv}_{\mathrm{ZK}}(\mathcal{B})$.

- $\mathbf{Game_2}$: This game is identical to $\mathbf{Game_1}$, except that the ciphertexts are replaced by encryptions of different values. An adversary $\mathcal{D}$ can distinguish between $\mathbf{Game_1}$ and $\mathbf{Game_2}$ with the advantage $\mathrm{Adv}_{\mathrm{CPA}}(\mathcal{D})$.

- $\mathbf{Game_3}$: In this game the way of computing $\mathbf{t}_0$ and $\mathbf{t}_1$ is different from the previous. Instead of sampling $\mathbf{x}_0$, $\mathbf{x}_1$ and computing $\mathbf{t}_0 = \mathbf{B}\mathbf{x}_0 + \mathrm{H}(\mathrm{com}_{\mathbf{x}_0}, \mu_b, \mathrm{ID})$, $\mathbf{t}_1 = \mathbf{B}\mathbf{x}_1 + \mathrm{H}(\mathrm{com}_{\mathbf{x}_1}, \mu_{b'}, \mathrm{ID})$, the challenger samples $\mathbf{u}_0$, $\mathbf{u}_1$ and sets $\mathbf{t}_0 = \mathbf{u}_0 + \mathrm{H}(\mathbf{c}_0, \mu_b, \mathrm{ID})$, $\mathbf{t}_1 = \mathbf{u}_1 + \mathrm{H}(\mathbf{c}_1, \mu_{b'}, \mathrm{ID})$. The vectors $\mathbf{c}_0$ and $\mathbf{c}_1$ are uniformly sampled from the Ajtai commitment space. By the leftover hash lemma (Lemma 4), $\mathbf{B}\mathbf{x}_0$ and $\mathbf{B}\mathbf{x}_1$ are statistically indistinguishable from the vectors $\mathbf{u}_0$ and $\mathbf{u}_1$. Moreover, $\mathbf{c}_0$ and $\mathbf{c}_1$ are indistinguishable from $\mathrm{com}_{\mathbf{x}_0}$ and $\mathrm{com}_{\mathbf{x}_1}$ by the computational hiding property of the Ajtai commitment scheme, except with the advantage $\mathrm{Adv}_{\mathrm{HIDE}}(\mathcal{D})$.

□

4.2 Concrete Instantiation

In this section, we provide a theoretical analysis of our scheme in Fig. 3, and examine the size of the resulting signature. The main components of our scheme are : (i) SHA-2 as the hash function H, (ii) the GPV signature scheme on modules from [BEP+21] with necessary modifications (iii) for the NIZK proof π_2 we follow the protocol from [LNP22, Figure 10]. Since we finally publish the proof π_2 along with the commitment $\mathrm{com}_{\mathbf{x}}$, the size of our final signature mostly depends on our proof size, so our main interest is to make the proof generation efficient. The modulus $\mathbf{q}_{zk}$ of the proof system is chosen to be a product of 5 modulo 8 primes as specified by [LNP22, Lemma 2.5]. Accordingly, we begin by choosing a 5 modulo 8 prime q_s for the underlying signature scheme and another such q_1 and set $q_{zk} = q_s \cdot q_1$.

GPV signature scheme. We instantiate the scheme over the ring $\mathcal{R}_{256} = \mathbb{Z}_{q_s}[x]/\langle x^{256}+1\rangle$ with $q_s = 536871029$. The prime q_s is 5 modulo 8 and is of 30-bit. It is chosen in line with the work [BEP+21]. The TrapGen algorithm generates the public matrix $\mathbf{A}$ along with its $\mathbf{G}$-trapdoor $\mathbf{R}$ with standard deviation $\sigma > \eta_\epsilon(\mathbb{Z}^{2\mathrm{nd}})$ so that $\mathbf{A}$ is uniformly random under decisional MLWE assumption [Theorem 2]. The DelTrap algorithm internally uses SamplePre to sample each column of $\mathbf{R}_{\mathrm{ID}}$ which samples from a SIS lattice with standard deviation $s > \sqrt{(\alpha^2+1)s_1^2(\mathbf{R}) + \eta_\epsilon^2(\mathbb{Z}^{nm})}$ such that $\|\mathbf{R}_{\mathrm{ID}}^{(i)}\| \leq 12s\sqrt{mn}$. The SamplePre algorithm samples the

variable	description	instantiation
ρ	no. of equations to prove	0
ρ_{eval}	no. of evaluations with const. coef. zero	0
v_e	no. of exact norm proofs	3
v_d	no. of non-exact norm proofs	0
$\mathbf{s}_1$	committed message in the Ajtai part	$\mathbf{y}\|\mathbf{x}\|\mathbf{r}$
m	committed message in the BDLOP part	no msg.
$\mathbf{E}_1$	public matrix for $\|\mathbf{E}_1\mathbf{s} - \mathbf{v}_1\| \leq \beta_1$	$[\mathbf{I}_{m+dk}\mathbf{0}\mathbf{0}]$
$\mathbf{v}_1$	public vector for proving $\|\mathbf{E}_1\mathbf{s} - \mathbf{v}_1\| \leq \beta_1$	$\mathbf{0}$
β_1	upper-bound on $\|\mathbf{E}_1\mathbf{s} - \mathbf{v}_1\| \leq \beta_1$	β_y
$\mathbf{E}_2$	public matrix for $\|\mathbf{E}_2\mathbf{s} - \mathbf{v}_2\| \leq \beta_2$	$[\mathbf{I}_m\mathbf{0}\mathbf{0}]$
$\mathbf{v}_2$	public vector for proving $\|\mathbf{E}_2\mathbf{s} - \mathbf{v}_2\| \leq \beta_2$	$\mathbf{0}$
β_2	upper-bound on $\|\mathbf{E}_2\mathbf{s} - \mathbf{v}_2\| \leq \beta_2$	β_x
$\mathbf{E}_3$	public matrix for $\|\mathbf{E}_3\mathbf{s} - \mathbf{v}_3\| \leq \beta_3$	$[\mathbf{1}\mathbf{0}\mathbf{0}]$
$\mathbf{v}_3$	public vector for proving $\|\mathbf{E}_3\mathbf{s} - \mathbf{v}_3\| \leq \beta_3$	$\mathbf{0}$
β_3	upper-bound on $\|\mathbf{E}_3\mathbf{s} - \mathbf{v}_3\| \leq \beta_3$	β_r

Table 1: Instantiating proof π_2

signature $\mathbf{y}$ within ℓ_2 norm bound $12s'\sqrt{nm+ndk}$, with $s' > \sqrt{(\alpha^2+1)s_1^2(\mathbf{R}_{\text{ID}}) + \eta_\epsilon^2(\mathbb{Z}^{nm+ndk})}$ as a standard deviation. Vector $\mathbf{y}$ is sampled from the SIS lattice $\{\mathbf{y} : \mathbf{A}_{\text{ID}}\mathbf{y} = \mathbf{t} \bmod q_s\}$ using the $\mathbf{G}$-trapdoor $\mathbf{R}_{\text{ID}}$ of $\mathbf{A}_{\text{ID}}$.

Blinding the message. In order to blind the actual message from the signer, we sample a secret vector $\mathbf{x} \in \mathcal{R}_{256}^m$ to mask μ and send $\mathbf{t}$ to the signer for the signature. We also use a SHA-2 hash function, which takes as input μ , a commitment to $\mathbf{x}$ and the public ID of the signer. The vector $\mathbf{t} \in \mathcal{R}_{256}^d$ is already uniformly random here since we are using a one-time pad style hiding. So, $\mathbf{t}$ reveals nothing to the signer about μ and $\mathbf{t}$ must also not reveal $\mathbf{x}$ when the user publishes the final signature ($\text{com}_{\mathbf{x}}, \mu$ is public here). We choose the standard deviation σ_x of sampling each coefficient of $\mathbf{x}$ accordingly such that given $\mathbf{t} = \mathbf{B}\mathbf{x} + \text{H}(\text{com}_{\mathbf{x}}, \mu, \text{ID})$, finding $\mathbf{x}$ with $\|\mathbf{x}\| \leq \beta_{\mathbf{x}} = \sigma_{\mathbf{x}}\sqrt{mn_s}$ is hard. With the considerations and neglecting ct for now, we obtain the transcript size $(\mathbf{t}, \mathbf{y})$ to be around 82.54 KB.

Commitment Scheme. We generate $\text{com}_{\mathbf{x}} = \mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r}$ to the secret vector $\mathbf{x}$ using the Ajtai commitment scheme and use the commitment randomness $\mathbf{r}$. The $\text{com}_{\mathbf{x}}$ should be both hiding and binding. We make use of (Lemma 4) to set the dimensions of $\mathbf{A}_1, \mathbf{A}_2, \mathbf{r}$ and also the standard deviations with which each coefficient of $\mathbf{x}$ and $\mathbf{r}$ needs to be sampled such the $\text{com}_{\mathbf{x}}$ reveals no information about $\mathbf{x}$. The binding property reduces to solving

$$[\mathbf{A}_1|\mathbf{A}_2]\begin{pmatrix}\mathbf{x}\\ \mathbf{r}\end{pmatrix}, \|(\mathbf{x},\mathbf{r})^\top\| \leq \sqrt{\beta_{\mathbf{x}}^2 + \beta_{\mathbf{r}}^2}$$

Considering both these security requirements, we have set the commitment size to around 2.81 KB.

For our construction, we exploit the fact that one linear relation over $\mathcal{R}_{256}$ can be mapped to 2 linear relations over $\mathcal{R}_{128}$ [LNPS21, Section 2.8], thus all the linear relaions discussed in this section till now can be views as 2 relations over $\mathcal{R}_{128}$. This observation is relevant in the zero-knowledge proof discussion that follows, as we use the subring $\mathcal{R}_{128}$ for instantiating it.

Zero-knowledge proof. We follow the exact protocol from the paper [LNP22, Figure 10]. To construct the zero-knowledge proof π_2 in our scheme, we need to prove knowledge of $\mathbf{x} \in$

$\mathcal{R}_{128}^{2\cdot m}, \mathbf{y} \in \mathcal{R}_{128}^{2\cdot(m+dk)}, \mathbf{r} \in \mathcal{R}_{128}^{2\cdot 2}$ with small norms $\beta_x, \beta_y, \beta_r$ respectively such that,

$$\mathbf{A}_1\mathbf{x} + \mathbf{A}_2\mathbf{r} = \mathrm{com}_{\mathbf{x}} \text{ and } \mathbf{A}_{\mathrm{ID}}\mathbf{y} = \mathbf{B}\mathbf{x}_2 + \mathrm{H}(\mathrm{com}_{\mathbf{x}}, \mu, \mathrm{ID})$$

We commit to the vector $(\mathbf{y}|\mathbf{x}|\mathbf{r})$ and prove the above two linear relations involving this vector, along with the three ℓ_2 norm bounds $\|\mathbf{y}\| \leq \beta_y, \|\mathbf{x}\| \leq \beta_x, \|\mathbf{r}\| \leq \beta_r$. Following usual norms, we do not discuss the entire protocol but rather provide a sketch on how to instantiate [LNP22, Figure 10] in Table 1. Note that the linear relations that we need to proof incur negligible additional cost in terms of size [LNP22, Figure 4], so we choose to neglect that.

The final concrete parameters of our entire scheme have been derived considering all the above discussed security notions and are given in Table 2.The table is divided into two sets of parameters, one for the basic GPV signature scheme, commitment and blinding and the rest for the zero-knowledge proof. The notations followed by the second set of parameters are the ones used in the paper [LNP22] (except n_{zk}, q_{zk}).We have used the $q_{zk} = q_s \cdot q_1 = 536871029 \cdot 99327398173 = 53326002465031230017 \approx 2^{65}$ as the proof system modulus where both q_s and q_1 are 5 modulo 8 primes. The parameters κ, η follow from [LNP22, Fig.3] and we choose γ to be the largest such that the underlying MSIS problem is still hard and $D < \log_2(\gamma/\kappa d)+1$ [LNP22, Sec. 6.1]. Both γ and D are needed to apply Dilithium compression technique which further reduces the proof size by cutting the low order bits of the commitments. The values of $\gamma_1, \gamma_2, \gamma^{(e)}, \nu$ effect the expected number of repetitions while generating a valid signature. Here, we use the values in [AKSY22, Table 2] for $\gamma_1, \gamma_2, \gamma^{(e)}, \nu, \gamma, D$ and obtain a repetition rate of approximately 10.76 [LNP22, Sec. 6.1].One can even choose to calibrate them pertaining to the required security bits. The parameters m_2 and n are chosen such that the MSIS and MLWE problems that underlie the zero-knowledge protocol are hard, and $m_1 = 2 \cdot m + 2 \cdot (m + dk) + 2 + 3 = 569$ counts the length of the committed message $(\mathbf{y}|\mathbf{x}|\mathbf{r})$ as a vector over $\mathcal{R}_{128}$ (adding 3 as we have 3 norm equations) .

Using these parameters we have obtained a proof size of 327.84 KB and a overall signature size (π_2 and com) of 330.66 KB. We have generated the parameters following the security methodology from [ADPS15] for classical core-SVP hardness of about 111 bits and also adapted certain parameters for optimising the proof size from the paper [ESLR22]. Anyone can verify our parameters via a script available at `https://github.com/MrittikaNandi/round_optimal_IDBS.git`.

5 Conclusion

To the best of our knowledge, we have proposed the first concrete construction of a round optimal identity-based blind signature scheme under module lattice assumptions. The estimated signature size of our proposed scheme is approximately 330.66 KB. For certificateless communication and to get scalability, this type of work is important. In modern days scenario, centralised key generation might be harmful. To overcome this situation in our design, for thresholdising the msk among n parties, we can simply apply the techniques given in the paper [ALLW25] and left as a future work.

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parameters	description	values
λ	security parameter	128
q_s	modulus	536871029
k	bitlength of q_s	30
n_s	degree of the ring $\mathcal{R}_{q_s}$	256
d	dimension of the module	3
σ	standard deviation for TrapGen	71
s	standard deviation for DelTrap	4992
s'	standard deviation for SamplePre in sign	3297
α	standard deviation of G-sampling	89
t	gaussian tailcut	12
$\beta_{\mathbf{y}}$	norm bound on $\mathbf{y}$ $(ts'\sqrt{(m+dk)n_s})$	8633296
$\beta_{\mathbf{x}}$	norm bound on $\mathbf{x}$ $(60999\sqrt{mn_s})$	9562651
$\beta_{\mathbf{r}}$	norm bound on $\mathbf{r}$ $(195\sqrt{n_s})$	3120
q_{zk}	proof system modulus	$\approx 2^{65}$
n_{zk}	degree of the ring $\mathcal{R}_{q_{zk}}$	128
l	no. of factors $x^{n_{zk}}+1$ splits into modq_{zk}	2
γ_1	rej. samp. const. for masking $c\mathbf{s}_1$	10
γ_2	rej. samp. const. for masking $c\mathbf{s}_2$	1.5
$\gamma^{(e)}$	rej. samp. for l2 norm bound	3
κ	challenge space bound	2
η	const. for challenge space	59
ν	randomness $\mathbf{s}_2$ sampled from this bound	1
n	no. of rows in $\mathbf{A}_1, \mathbf{A}_2$ in ABDLOP	13
m_1	'Ajtai' commited secret belongs to $\mathcal{R}_q^{m_1}$	569
m_2	length of randomness $\mathbf{s}_2$	39
ℓ	'BDLOP' committed message belongs to $\mathcal{R}_q^{\ell}$	0
λ	no. of garbage commitments g_j	2
γ	parameter to cut low-order bits from $\mathbf{w}$	2^{24}
D	no. of low order bits cut from $\mathbf{t}_A$	16
	BKZ blocksize b to break LWE	389
	Classical security	113
	Quantum security	103
	BKZ blocksize b to break SIS	377
	Classical security	110
	Quantum security	100
	repetition rate	10.76
	proof size	327.84 KB
	signature size	330.66 KB

Table 2: Instantiating the scheme

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